# Left and right production matrices and total positivity of different types

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**Abstract**

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# Contents

# 1 Introduction

A (finite or infinite) matrix of real numbers is called ***totally positive*** (TP) if all its minors are nonnegative. A matrix of polynomials with real coefficients is called ***coefficientwise totally positive*** if all its minors are polynomials with nonnegative coefficients. Let $A = (a_{n,k})_{n,k\geq 0}$ be a lower-triangular matrix of nonnegative real numbers. In [10], some of us noticed that for such a matrix, we may always ask the following four questions:

(a) Is $A$ totally positive?

(b) Is the lower-triangular matrix $\overleftarrow{A} \stackrel{\text{def}}{=} (a_{n,n-k})_{n,k\geq 0}$, obtained by reversing the rows of $A$, totally positive? (Here $a_{n,n-k} \stackrel{\text{def}}{=} 0$ when $n < k$.)

(c) Let $A_n(x) = \sum_{k=0}^{n} a_{n,k}x^k$ denote the row-generating polynomial of the $n$-th row of $A$. Are the polynomials $A_n(x)$ negative-real-rooted?[1] This is equivalent [17, pp. 395, 399] to asking if the Toeplitz matrix of the $n$-th row sequence $(a_{n,k})_{k\geq 0}$ — that is, the matrix $(a_{n,i-j})_{i,j\geq 0}$ with $a_{n,k} \stackrel{\text{def}}{=} 0$ for $k < 0$ — is totally positive. In this situation we say that the $n$-th row sequence $(a_{n,k})_{k\geq 0}$ is ***Toeplitz-totally positive***.

(d) Is the Hankel matrix of the polynomial sequence $(A_n(x))_{n\geq 0}$ — that is, the matrix $(A_{i+j}(x))_{i,j\geq 0}$ — coefficientwise totally positive in the variable $x$? In this situation we say that the sequence $(A_n(x))_{n\geq 0}$ is ***coefficientwise Hankel-totally positive***.

For several important combinatorial triangles, the answers to all four of these questions have been proven to be yes: these include the matrix of binomial coefficients, the matrix of Stirling cycle numbers, the matrix of Stirling subset numbers; see [10] for details. Of these, the total positivity of the reversal of the Stirling subset numbers was only recently shown by us few years ago [6]. Next consider $A = \left(\left\langle {n \atop k} \right\rangle\right)_{n,k\geq 0}$, the matrix of Eulerian numbers, which count the permutations on $[n]$ with $k$ descents (or $k$ excedances). In this case, the answers to questions (c) and (d) is known to be true [16, 29]. However, here the total positivity of the matrix $A$ is equivalent to the total positivity of the matrix $\overleftarrow{A}$, and even though this was conjectured by Brenti in 1996 [5] the question is still open even after three decades.

A sufficient condition that has often been used to prove questions (a),(d) true simultaneously is the *method of production matrices* see for example [**?**]. On the other hand, one of us along with Fu and Ruan [7], provided a sufficient condition that implies the truth of questions (a), (b), (c) simultaneously: the existence of a *totally positive left-production matrix*.

[1] A polynomial with real coefficients is said to be ***real-rooted*** if it is either identically zero or all its (complex) zeros are real; it is said to be ***negative-real-rooted*** if it is either identically zero or all its (complex) zeros lie in $(-\infty, 0]$. Since we are here assuming that all the matrix entries $a_{n,k}$ are nonnegative, clearly $A_n(x)$ cannot have any *positive* real zeros unless it is identically zero. So negative-real-rootedness and real-rootedness are equivalent here.

The most general setting to consider the questions (a),(b),(c),(d) is to take a partially ordered commutative ring $R$ and let the entries of the matrix $A$ be $a_{nk} \in R$. The notions of total positivity, Hankel-total positivity and Toeplitz-total positivity are then defined in the obvious way. The main result of this paper is Theorem 2.3 which generalises [7, Theorem 1.3], which is in the setting when $R = \mathbb{R}$, to the setting of a partially ordered commutative ring; see Section 2.2. We will then write out the relation between the two notions of production matrices and left-production matrices in Section 2.3. In this paper, we will refer to production matrices as *right-production matrices.*

We remark that the proof of [7, Theorem 1.3], uses the fact that any invertible lower-triangular matrix with real entries has a planar digraph representation ( [7, Theorem 2.5]); this fact itself relies on a bidiagonal factorisation of such a matrix, see [17, Theorem 6.10]. However, such a bidiagonal factorisation may not exist if $R$ is not a field. Our proof of Theorem 2.3 is algebraic and works for any general partially ordered commutative ring $R$.

After developing the general theory, we will write out some applications of Theorem 2.3. Our first main application (Section 4) here is an algebraic proof of the total positivity of the *ace*-matrix in [6, Theorem 1.5] that we had promised five years ago. We will also show that the row-generating Toeplitz matrix is totally positive.

Our second application will be some results concerning the multivariate Laguerre coefficient matrix [9], see Section 5.1. Our final application (Section 6) is a proof of the total positivity of the matrix of linearly ordered phylogenetic trees [10, Conj. 4.17].

Finally, after raising questions (a),(b),(c),(d), one wonders if these have any relations or implications among each other. We will visit this in Section 7 where we construct an example for each of the $2^4 = 16$ different truth table values to the four questions. This will show that there is no relation among these four questions in general.

# 2 Left production matrices

## 2.1 Definition of left production matrices

Let $Q$ be a row-finite or column-finite matrix. Define the following matrix $\widehat{A}$

$$\widehat{A} \;=\; Q \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & Q \end{array}\right] \left[\begin{array}{c|c} I_2 & \mathbf{0} \\ \hline \mathbf{0} & Q \end{array}\right] \left[\begin{array}{c|c} I_3 & \mathbf{0} \\ \hline \mathbf{0} & Q \end{array}\right] \cdots \quad . \tag{2.1}$$

Note that (2.1) can equivalently be rewritten as

$$\widehat{A} \;=\; Q \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widehat{A} \end{array}\right] \tag{2.2}$$

since (2.1) imply (2.2) by substitution on the right-hand side, while (2.2) imply (2.1) by iteration. We call $Q$ the ***left-production matrix***, and $\widehat{A}$ the ***left-output ma-***

***trix***. We may rewrite (2.2) as the recurrence

$$\widehat{a}_{n0} = q_{n0} \tag{2.3a}$$

$$\widehat{a}_{n,k+1} = \sum_{i=0}^{\infty} q_{n,i+1}\widehat{a}_{i,k} \quad \text{for } n,k \geq 0 \tag{2.3b}$$

Fully carrying out the recurrence, we have

$$\widehat{a}_{nk} = \sum_{i_1,\ldots,i_k} q_{n,i_1+1}q_{i_1,i_2+1}\cdots q_{i_{k-1},i_k+1}q_{i_k,0} \,. \tag{2.4}$$

If we further assume that $Q$ is lower-Hessenberg, we get the recurrence

$$\widehat{a}_{n,k+1} = \sum_{i=0}^{n} q_{n,i+1}\widehat{a}_{i,k} \quad \text{for } n,k \geq 0 \tag{2.5}$$

with initial condition $\widehat{a}_{n0} = q_{n0}$. In this case, the matrix $\widehat{A}$ is a dense matrix with zeroth row given by $\widehat{a}_{0k} = q_{00}\, q_{01}^k$.

Finally, if we assume that $Q$ is lower-triangular, than the matrix $A$ is also lower triangular and we get the recurrence

$$\widehat{a}_{n+1,k+1} = \sum_{i=0}^{n} q_{n+1,i+1}\widehat{a}_{i,k} \quad \text{for } n,k \geq 0 \tag{2.6}$$

with initial condition $\widehat{a}_{n0} = q_{n0}$. The diagonal elements of $\widehat{A}$ are partial products of those of $Q$:

$$\widehat{a}_{nn} = \prod_{i=0}^{n} q_{ii} \,. \tag{2.7}$$

In particular, $\widehat{A}$ is invertible $\iff$ the diagonal elements of $\widehat{A}$ are invertible $\iff$ the diagonal elements of $Q$ are invertible $\iff$ $Q$ is invertible. And in this case, $Q$ can be recovered from $\widehat{A}$ by

$$Q = \widehat{A}\cdot \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widehat{A}^{-1} \end{array}\right] = \widehat{A}\left(\mathbf{e}_{00} + \Delta^{\mathrm{T}}\widehat{A}\Delta\right)^{-1} \tag{2.8}$$

where $\Delta$ is the matrix with 1 on the superdiagonal and 0 elsewhere.

**GIVE SOME HISTORY!!!!!!**

We now write the following proposition.

**Proposition 2.1.** *Let $Q$ be a row-finite matrix with entries in a commutative ring $R$, with left-output matrix $\widehat{A}$; and let $B$ be an invertible row-finite matrix such that $B^{-1}$ is also row-finite. Then the matrix $B\widehat{A}$ is the left-output matrix of the following matrix*

$$B\,Q\left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & B^{-1} \end{array}\right] \,. \tag{2.9}$$

Proof. Let $Q_2$ be the matrix in (2.9). We write out (2.1) with the matrix $Q_2$ and see that it gives us $BA$. $\square$

### 2.1.1 Combinatorial interpretation

Let $Q = (q_{ij})_{i,j \geq 0}$ be a lower-triangular matrix and let $\widehat{A} = (\widehat{a}_{nk})_{n,k\geq 0}$ be its left-output matrix. We now provide a combinatorial interpration of the matrix entries of $\widehat{A}$.

Consider the alphabet $\mathbb{N} \cup \{|\}$ and set $w_n$ to be the word $w_n = 0\,1\,2\,\cdots\,n$ on this alphabet. Let $\mathbb{B}_n$ be the set of all words obtained by inserting bars "|" between any two consecutive letters of $w_n$. For example $\mathbb{B}_2 = \{0\,1\,2, 0|1\,2, 0\,1|2, 0|1|2\}$. Clearly, the set $\mathbb{B}_n$ is in bijection with subsets of the set $[n]$ and therefore has cardinality $2^n$: a subset $S \subseteq [n]$ encodes the letters occuring before bars in the word $w \in \mathbb{B}_n$. We use $\mathbb{B}_{n,k} \subseteq \mathbb{B}$ to denote the subcollection of words each containing $k$ bars.

To each finite interval $[i,j] \subset \mathbb{N}$, we assign the weight $\mathrm{wt}([i,j]) = q_{ji}$. The weight of a word $w \in \mathbb{B}_{n,k}$, denoted by $\mathrm{wt}(w)$ is the product of the weights of the $(k+1)$ subintervals of $[0,n]$ separated by the bars in $w$. For example, the words in $\mathbb{B}_2$ have the following weights:

$$\mathrm{wt}(0\,1\,2) \;=\; q_{2,0} \quad \mathrm{wt}(0|1\,2) \;=\; q_{0,0}q_{2,1} \quad \mathrm{wt}(0\,1|2) \;=\; q_{1,0}q_{2,2} \quad \mathrm{wt}(0|1|2) \;=\; q_{0,0}q_{1,1}q_{2,2}\,.$$

Finally, we define $\mathrm{wt}(\mathbb{B}_{n,k}) := \sum_{w\in\mathbb{B}_{n,k}} \mathrm{wt}(w)$. For example, $\mathrm{wt}(\mathbb{B}_{2,1}) = q_{0,0}q_{2,1} + q_{1,0}q_{2,2}$.

We are now ready to state our combinatorial interpretation of the left-output matrix $\widehat{A}$.

**Proposition 2.2.** *For a lower triangular left-production matrix $Q = (q_{ij})_{i,j\geq 0}$ with corresponding left-output matrix $\widehat{A} = (\widehat{a}_{nk})_{n,k\geq 0}$, the entry $\widehat{a}_{n,k}$, is given by the identity:*

$$\widehat{a}_{n,k} \;=\; \mathrm{wt}(\mathbb{B}_{n,k})\,. \tag{2.10}$$

Proof. First notice that $\mathrm{wt}(\mathbb{B}_{n,0}) = (\{w_n\}) = q_{n0}$; thus the initial condition is satisfied. It is easy to see that the quantities $\mathrm{wt}(\mathbb{B}_{n,k})$ satisfy the recurrence (2.6). We omit the details. $\square$

## 2.2 Left production matrices and total positivity

We now state the main result of this paper.

**Theorem 2.3.** *Let $Q = (q_{ij})_{i,j\geq 0}$ be a row-finite or a column-finite left-production matrix with elements in a partially ordered commutative ring $R$, let $\widehat{A} = (\widehat{a}_{nk})_{n,k\geq 0}$ be its left-output matrix. Let us assume that $Q$ is totally positive of order $r$. We then have*

- *(a) The matrix $\widehat{A}$ is also totally positive of order $r$.*
- *(b) If $Q$ is lower triangular (so that the matrix $\widehat{A}$ is lower triangular), then the reversal $\overleftarrow{\widehat{A}}$ exists and it is totally positive of order $r$.*
- *(c) Let $Q$ be either lower-triangular, or more generally, lower-Hessenberg. For any fixed row $n \geq 0$, the infinite lower triangular Toeplitz matrix $T_\infty((\widehat{a}_{n,k})_{k\geq 0}) = (\widehat{a}_{n,i-j})_{i,j\geq 0}$ is Toeplitz-totally positive of order $r$.*

This is a striking result. **SAY WHY???????**

**Remark.** Theorem 2.3 was proved by one of us in the setting when $R = \mathbb{R}$, and for total positivity of order $\infty$ and also $Q$ was assumed to be lower triangular; see [7, Theorem 1.3]. The proof in this case is via the existence of a planar network representation of the matrix $Q$ ( [7, Theorem 2.5]) which itself relies on a bidiagonal factorisation of a totally positive matrix with real entries, see [17, Theorem 6.10].

However, a bidiagonal factorisation may not exist if $R$ is not a field. Our proof of here is algebraic and works for a general partially ordered commutative ring. ■

In preparation for the proof of Theorem 2.3, let us introduce some useful notation. For any infinite matrix $X = (x_{ij})_{i,j\geq 0}$ and any $M, N \in \mathbb{N}\cup\{\infty\}$, let us write $X_{M\times N} = (x_{ij})_{0\leq i\leq M-1, 0\leq j\leq N-1}$ for its $M \times N$ leading principal submatrix. When $M = N$, we use $X_N$ instead of $X_{M\times N}$. Then observe that the identity (2.2) for infinite matrices is equivalent to the sequence of identities

$$\widehat{A}_{M\times(N+1)} \;=\; Q_{M\times\infty} \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widehat{A}_{\infty\times N} \end{array}\right] \tag{2.11}$$

for all its leading principal submatrices. In particular, when $Q$ is lower-Hessenberg the following identities suffice

$$\widehat{A}_{(N+1)\times(N+2)} \;=\; Q_{(N+1)\times(N+2)} \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widehat{A}_{(N+1)} \end{array}\right], \tag{2.12}$$

and if $Q$ is lower-triangular, we may further simplify to

$$\widehat{A}_{N+1} \;=\; Q_{N+1} \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widehat{A}_{N} \end{array}\right]. \tag{2.13}$$

Proof of Theorem 2.3. *(a)* This is immediate from the definition in (2.1).

*(b)* Here the matrix $Q$ is lower triangular. Define an $N\times N$ matrix $B_{/N} = (b_{ij}^{(N)})_{0\leq i,j\leq N-1}$ by

$$b_{ij}^{(N)} \;=\; \widehat{a}_{N-1-i,j-i}\;. \tag{2.14}$$

That is, the rows of $B_{/N}$ are the rows of $A_N$ taken in reverse order (i.e. interchanging up with down), right-justified in a matrix of $N$ columns. Note in particular that the top row of $B_{/N}$ is the same as the bottom row of $A_N$, and that

$$B_{/N+1} \;=\; \left[\begin{array}{c|c} \multicolumn{2}{c}{\widehat{\boldsymbol{a}}_N} \\ \hline \mathbf{0} & B_{/N} \end{array}\right] \tag{2.15}$$

where $\widehat{\boldsymbol{a}}_N \stackrel{\rm def}{=} (\widehat{a}_{Nj})_{0\le j\le N}$ = bottom row of $\widehat{A}_{N+1}$ = top row of $B_{/N+1}$. We then have

$$\left[\begin{array}{c} \widehat{A}_{N+1} \\ \hline B_{/N+1} \end{array}\right] = \left[\begin{array}{c|c} Q_{N+1} & \\ \hline \boldsymbol{q}_N & \\ \hline & I_N \end{array}\right] \left[\begin{array}{c|c} 1 & \\ \hline & \widehat{A}_N \\ \hline & B_{/N} \end{array}\right] \tag{2.16}$$

where the blank entries are zero, and $\boldsymbol{q}_N \stackrel{\rm def}{=} (q_{N,j})_{0\le j\le N}$ = bottom row of $Q_{N+1}$. It follows by induction that if $Q$ is $\mathrm{TP}_r$, then so is $\left[\begin{array}{c} \widehat{A}_N \\ \hline B_{/N} \end{array}\right]$ for all $N \ge 0$, and hence so are $B_{/N}$ for all $N \ge 0$.

Now let $R_N = (\delta_{i+j,N-1})_{0\le i,j\le N-1}$ be the $N\times N$ matrix with 1 on the antidiagonal and 0 elsewhere. Left-multiplication (resp. right-multiplication) of an $N \times N$ matrix by $R_N$ reverses the order of its rows (resp. columns). In particular, when $B_{/N}$ is $\mathrm{TP}_r$, then so is $R_N B_{/N} R_N$, which has matrix elements

$$(R_N B_{/N} R_N)_{ij} \;=\; b^{(N)}_{N-1-i,N-1-j} \;=\; \widehat{a}_{i,i-j} \;=\; (\overleftarrow{\widehat{A}})_{ij} \tag{2.17}$$

for $0 \le i,j \le N-1$. Therefore, if $Q$ is $\mathrm{TP}_r$, then so is $(\overleftarrow{\widehat{A}})_N$ for all $N \ge 0$, and hence so is $\overleftarrow{\widehat{A}}$.

*(c)* Here we assume $Q$ to be lower-Hessenberg. First consider the matrix $\left[\begin{array}{c|c} \multicolumn{2}{c}{Q_{N\times(N+R)}} \\ \hline \mathbf{0} & I_R \end{array}\right]$; the block $I_R$ here starts at column $N+1$ which is the last column until which $Q_{N\times(N+R)}$ has a non-zero entry. If $Q$ is totally positive of order $r$, so is this matrix.

Now consider the following matrix

$$M_{N,R} \stackrel{\rm def}{=} \left[\begin{array}{c|c} \multicolumn{2}{c}{Q_{N\times(N+R)}} \\ \hline \mathbf{0} & I_R \end{array}\right] \left[\begin{array}{c|cc} I_1 & & \\ \hline & \multicolumn{2}{c}{Q_{N\times(N+R-1)}} \\ \hline & \mathbf{0} & I_{R-1} \end{array}\right] \left[\begin{array}{c|cc} I_2 & & \\ \hline & \multicolumn{2}{c}{Q_{N\times(N+R-2)}} \\ \hline & \mathbf{0} & I_{R-2} \end{array}\right] \cdots \left[\begin{array}{c|c} I_R & \\ \hline & Q_{N\times(N+1)} \end{array}\right]. \tag{2.18}$$

In this product, each individual matrix has shape $(N+R)\times(N+R)$ except the rightmost one which has shape $(N+R)\times(N+R+1)$.

We claim that the following identity holds:

$$\big(T_\infty((\widehat{a}_{N-1,k})_{k\ge 0})^{\rm T}\big)_{R+1} = M_{N,R}\left[\begin{array}{c} N, N+1, \ldots, N+R \\ 0, 1, \ldots, R \end{array}\right], \tag{2.19}$$

where the right-hand side denotes the bottom left $(R+1)\times(R+1)$ submatrix of $M_{N,R}$. Note that (2.19) is equivalent to showing that for all $i,j$ satisfying $N \le i \le N+R$, and $0 \le j \le R$, the entry $(M_{N,R})_{i,i-N+j} = \widehat{a}_{N-1,j}$.

We use (2.18) to compute the entry $i, i+j$ of the matrix $M_{N,R}$. Notice that in this computation, we may ignore the first $i-N$ factors on the right-hand side of (2.18) as the lower right portion below the entry $i,i$ is just the identity matrix. This gives us

$$
\begin{aligned}
(M_{N,R})_{i,i-N+j} &= \left( \left[ \begin{array}{c|cc} I_{i-N} & & \\ \hline & & \\ & \multicolumn{2}{c}{Q_{N\times(2N+R-i)}} \\ \hline & \mathbf{0} & I_{R+N-i} \end{array} \right] \cdots \left[ \begin{array}{c|c} I_R & \\ \hline & Q_{N\times(N+1)} \end{array} \right] \right)_{i,i-N+j} && \text{(2.20a)} \\
&= \left( \left[ \begin{array}{cc} \multicolumn{2}{c}{Q_{N\times(2N+R-i)}} \\ \hline \mathbf{0} & \multicolumn{1}{|c}{I_{R+N-i}} \end{array} \right] \cdots \left[ \begin{array}{c|c} I_{R+N-i} & \\ \hline & Q_{N\times(N+1)} \end{array} \right] \right)_{N,j} && \text{(2.20b)} \\
&= \widehat{a}_{N-1,j}\,. && \text{(2.20c)}
\end{aligned}
$$

where the second equality is obtained by taking the lower right $(R+2N-i)\times(R+2N-i)$ submatrix of each individual factor except for the last one where we take the lower-right $(R+2N-i)\times(R+2N-i+1)$. The third equality is simply (2.1). This establishes the claimed identity (2.19).

Finally, if $Q$ is $\mathrm{TP}_r$, so is $M_{N,R}$ for every $N, R$ and so must be the Toeplitz matrix $T_\infty((\widehat{a}_{N-1,k})_{k\geq 0})$. $\square$

**Remark.** The working of Equation (2.20) is inspired by the combinatorial arguments used in the proof of [7, Theorem 1.4]. Notice that the their theorem requires the left production matrix to be lower triangular. We have lifted this requirement on all of the statements except for (b). $\blacksquare$

**PLEASE CHECK IF I HAVE DONE THE PROOF CORRECTLY!!!!!!!!**

## 2.3 Relation between left and right production matrices

The method of production matrices [12, 13] has become in recent years an important tool in enumerative combinatorics. This method has also been successfully employed to obtain several total positivity results, see for example, **CITE!!!!!!!**. In this section we briefly recall production matrices following [26, sections 2.2 and 2.3] and then show their relation to our left-production matrices. To avoid confusion with left-production matrices, we will refer to them as right-production matrices in this paper.

Let $P = (p_{ij})_{i,j\geq 0}$ be an infinite matrix with entries in a commutative ring $R$, and assume that $P$ is either row-finite or column-finite (so that powers of $P$ are well-defined). Now define a matrix $A = (a_{nk})_{n,k\geq 0}$ by

$$a_{nk} \;=\; (P^n)_{0k} \tag{2.21}$$

(note in particular that $a_{0k} = \delta_{k0}$). We call $P$ the ***right-production matrix*** and $A$ the ***right-output matrix***, and we write $A = \mathcal{O}(P)$. Another equivalent formulation of (2.21), is the following recurrence

$$a_{nk} \;=\; \sum_{i=0}^{\infty} a_{n-1,i}p_{ik} \;, \tag{2.22}$$

with the initial condition $a_{0k} = \delta_{k0}$. This is equivalent to $\Delta A = AP$, where recall that $\Delta$ is the matrix with 1 on the superdiagonal and 0 elsewhere. Note that if $P$ is lower-Hessenberg (i.e. vanishes above the first superdiagonal), then $\mathcal{O}(P)$ is lower-triangular; this is the most common case, and will be the case here. Conversely, when $A$ is lower-triangular with invertible diagonal entries and $a_{00} = 1$, then there exists a unique production matrix generating $A$, and it is obtained as

$$P \;=\; A^{-1}\Delta A \,. \tag{2.23}$$

Let us define the ***augmented right-production matrix*** to be the

$$\widetilde{P} \;\stackrel{\rm def}{=}\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \multicolumn{2}{c}{P} \end{array}\right] . \tag{2.24}$$

The relation $\Delta A = AP$ can be rewritten in terms of $\widetilde{P}$ as

$$A \;=\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & A \end{array}\right] \widetilde{P} \tag{2.25}$$

Iterating this, we can express the right-output matrix $A$ in a form that is similar to equation (2.1):

$$A \;=\; \cdots \left[\begin{array}{c|c} I_3 & \mathbf{0} \\ \hline \mathbf{0} & \widetilde{P} \end{array}\right] \left[\begin{array}{c|c} I_2 & \mathbf{0} \\ \hline \mathbf{0} & \widetilde{P} \end{array}\right] \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widetilde{P} \end{array}\right] \widetilde{P} \;. \tag{2.26}$$

Equation (2.26) justifies why we call $P$ as the right-production matrix of $A$.

Finally, we have the following relation between the left and right production matrices.

**Proposition 2.4.** *Let $P$ be a right-production matrix and let $A$ be its right-output matrix. Then the matrix $\widetilde{P}^{\rm T}$ is a left-production matrix and its left-output matrix is $A^{\rm T}$.*

Proof. We tranpose both sides of equation (2.26). □

Thus, the left-production matrix of a matrix $A$ is the transpose of the right-production matrix of its transpose $A^{\rm T}$.

We finally end this section by stating an extremely shocking corollary.

**Corollary 2.5.** *Let $P$ be an upper triangular matrix and $A = \mathcal{O}(P)$ be its right output matrix. If $P$ is totally positive of order $r$, then each column sequence of $A$ is Toeplitz totally positive of order $r$.*

Proof. Consider the augmented right-production matrix $\widetilde{P}$ given by (2.25); its transpose $\widetilde{P}^{\mathrm{T}}$ is lower Hessenberg. The left output matrix of $\widetilde{P}^{\mathrm{T}}$ is $A^{\mathrm{T}}$ by Proposition 2.4. The proof now follows from Theorem 2.3(c). $\square$

We now state why this corollary is shocking. By virtue of [22, Theorem 9.7], we know that if a right production matrix (that is row-finite or column-finite) is totally positive, then its zeroth column is Hankel totally positive. However, we just noticed in Corollary 2.5 that for an upper triangular right production matrix, not only its zeroth column, but all columns are Toeplitz totally positive. However, on further inspection we can notice that for such a right production matrix, the column 0 of its right output matrix is simply $(p_{00}^n)_{n\geq 0}$ which is indeed both Hankel and Toeplitz totally positive.

# 3 Generalizations of some results of Bränden–Saud Maia Leite using the theory of left and right production matrices

In this section, we will study some notions developed by Bränden and Saud Maia Leite recently in [3] using our theory of left and right production matrices. We will work over an arbitrary partially ordered commutative ring and will thus be able to provide vast generalizations of their results [3, Theorems 3.7 and 4.3]. In Section 3.1 we study some sequences generated by a totally positive lower triangular matrix which turn out to be Toeplitz totally positive; in particular we generalize [3, Theorem 4.3]. Then in Section 3.2 we introduce $h$-polynomials and recall *chain polynomials* introduced by Bränden and Saud Maia Leite and then generalize their [3, Theorem 3.7].

## 3.1 Some Toeplitz totally positive sequences generated from a totally positive lower triangular matrix

Let $R$ be a lower-triangular matrix with entries in a partially ordered commutative ring. For any $i, j \in \mathbb{N}$ satisfying $i \geq j$, we consider the sequence $\big((R^k)_{ij}\big)_{k\geq 0}$. We first recall [3, Theorem 4.3(b)]:

**Theorem 3.1** ( [3, Theorem 4.3(b)]). *Let $R$ be a unit lower-triangular matrix of real numbers. If $R$ is totally positive then for each given $i, j$ satisfying $i \geq j$, the sequence $\big((R^k)_{ij}\big)_{k\geq 0}$ is Toeplitz totally positive.*

This result was proved using the interlacing of zeroes of chain polynomials (we will see these in the next subsection). We will now generalize their result: we will lift the restriction that the diagonal elements need to be 1 and our result will hold over any partially ordered commutative ring and for total positivity of any order. Our proof will be algebraic.

**Theorem 3.2.** *Let $R = (r_{n,k})_{n,k\geq 0}$ be a lower triangular matrix with elements in a partially ordered commutative ring $R$, and let $i, j \in \mathbb{N}$ be fixed satisfying $i \geq j$. Then* *if $R$ is totally positive of order $r$, the sequence $\big((R^k)_{ij}\big)_{k\ge 0}$ is Toeplitz totally positive of order $r$.*

Before proving this theorem, we first define the matrix $B := (b_{nk})_{n,k\ge 0}$ by

$$b_{nk} \;=\; (R^k)_{n,0}\;. \tag{3.1}$$

We then establish the following lemma.

**Lemma 3.3.** *Given a row-finite or column-finite matrix $R = (r_{nk})_{n,k\ge 0}$, we set $\widehat{R}$ to be the following matrix $\widehat{R} := (\mathbf{e}_{00} + R\Delta)$, i.e.,*

$$\widehat{R} \;=\; \left[\begin{array}{c|c} 1 & R \\ \hline & \\ \mathbf{0} & \end{array}\right]. \tag{3.2}$$

*Then $\widehat{R}$ when considered as a left-production matrix gives $B = \big((R^k)_{n,0}\big)_{n,k\ge 0}$ as its left output matrix.*

Proof. We show that the entries $b_{n,k}$ satisfy the recurrence (2.3). First notice that by definition $b_{n0} = \delta_{n0} = \big(\widehat{R}\big)_{n0}$ which is the initial condition that we want. Finally, we notice that

$$b_{n,k+1} \;=\; (R\cdot R^k)_{n,0} \;=\; \sum_i r_{ni}\, b_{i,k} \;=\; \sum_{i=0}^{\infty} \big(\widehat{R}\big)_{n,i+1}\, b_{i,k} \tag{3.3}$$

which is what we want. □

We are now ready to prove Theorem 3.2.

Proof of Theorem 3.2. If $R$ is totally positive of order $r$, so is $\widehat{R}$. When $j > 0$, we follow Brändén and Saud Maia Leite and recall their observation $(R^k)_{ij} = (S^k)_{i-j,0}$ where $S$ is the matrix obtained from $R$ by deleting the $j$ initial rows and columns of $R$; see [3, proof of Theorem 4.3]. Thus, it suffices to only deal with $j = 0$. Here, the desired result follows by using Lemma 3.3 together with Theorem 2.3(c) when applied to row $i$ of the matrix $B$. □

## 3.2 $h$-polynomials and chain polynomials

### 3.2.1 $h$-polynomials

Let $Q = (q_{nk})_{n,k\ge 0}$ be a lower-Hessenberg left production matrix and let $\widehat{A} = (\widehat{a}_{nk})_{n,k\ge 0}$ be its corresponding left output matrix. We recall that if the super-diagonal entries of $(q_{n,n+1})_{n\ge 0}$ are non-zero, then $\widehat{A}$ will be a dense matrix. Let $\widehat{A}_n(x) := \sum_{k=0}^{\infty} \widehat{a}_{nk}\, x^k$ denote the $n$-th row-generating o.g.f. of the matrix $\widehat{A}$.

We now rewrite the recurrence (2.5) in terms of $\widehat{A}_n(x)$ to obtain

$$\widehat{A}_0(x) \;=\; \frac{q_{00}}{1 \,-\, q_{01}x} \tag{3.4a}$$

$$\widehat{A}_{n+1}(x) \;=\; q_{n+1,0} \,+\, x \sum_{i=0}^{n+1} q_{n+1,i+1}\, \widehat{A}_i(x) \qquad \text{for } n \ge 0\;. \tag{3.4b}$$

From this rewriting we immediately get the following proposition.

**Proposition 3.4.** *Let $Q = (q_{nk})_{n,k\ge 0}$ be a lower-Hessenberg left production matrix and let $\widehat{A} = (\widehat{a}_{nk})_{n,k\ge 0}$ be its corresponding left output matrix. There exists a unique sequence of polynomials $(h_n(x))_{n\ge 0}$ such that the degree of the polynomial $h_n(x)$ is at most $n$, and the $n$-th row-generating o.g.f. of the matrix $\widehat{A}$ satisfies*

$$\widehat{A}_n(x) \;=\; \frac{h_n(x)}{\prod_{i=0}^{n}(1 \,-\, xq_{i,i+1})}\;. \tag{3.5}$$

The proof of Proposition 3.4 is straightforward from the recurrence (3.4).

**Definition 3.5.** *For a given lower-Hessenberg left production matrix $Q$, its corresponding* ***sequence of h-polynomimals*** *is the sequence $(h_n(x))_{n\ge 0}$ given by Proposition 3.4.*

An immediate corollary of the existence of $h$-polynomials is the following generalization of [3, Theorem 4.3(a)].

**Corollary 3.6.** *Let the underlying ring $R$ be a commutative ring with unity that is a $\mathbb{Q}$-algebra. Let $Q = (q_{n,k})_{n,k\ge 0}$ be a unit lower-Hessenberg matrix and let $\widehat{A} = (\widehat{a}_{nk})_{n,k\ge 0}$ be its corresponding left output matrix. We then have:*

*(a) For each $n \in \mathbb{N}$, there exists a polynomial $P_n : \mathbb{N} \to R$ of degree $n$ such that entries of the $n$-th row of $\widehat{A}$ are obtained by evaluating this polynomial, i.e. $P_n(k) = \widehat{a}_{n,k}$.*

*(b) Let $i, j \in \mathbb{N}$ be fixed satisfying $i \ge j$, and let $R = (r_{n,k})_{n,k\ge 0}$ be a unit lower-triangular matrix. The sequence $\big((R^k)_{i,j}\big)_{k\ge 0}$ is obtained by evaluating a polynomial of degree $i - j$.*

Proof. *(a)* From the existence of $h$-polynomials in Proposition 3.4 and the fact that the super diagonal entries are $q_{i,i+1} = 1$ for all $i \ge 0$, we have $\widehat{A}_n(x) \;=\; \dfrac{h_n(x)}{(1 \,-\, x)^{n+1}}$. Thus, from a well-known result **FIND CORRECT REFERENCE FOR THE MOST GENERAL RING!!!!!**, we get that the $n$-th row sequence is obtained by evaluating a polynomial $P_n : \mathbb{N} \to R$ of degree at most $n$. Finally, we will soon see in Lemma 3.7/Proposition 3.8 that $h_n(1) \ne 0$ which shows that $P_n$ has degree exactly $n$.

*(b)* This follows by combining (a) with Lemma 3.3. □

**Remark.** If a polynomial $f$ of degree $d$ satisfies $\sum_{n=0}^{\infty} f(n)x^n = \dfrac{g(x)}{(1-x)^{d+1}}$ for some polynomial $g$ of degree $d$, then the polynomial $g(x)$ is frequently called the $f$-Eulerian polynomial, see [18] and references therein. ■

Our first observation is that the coefficients of $h$-polynomials can be expressed in terms of minors of the matrix left production matrix $Q$. Here, for a finite set $S \subset \mathbb{N}$, we use the notation $S + a := \{s + a | s \in S\}$.

**Lemma 3.7.** *Let $Q = (q_{n,k})_{n,k\geq 0}$ be a unit lower-Hessenberg left production matrix and let $(h_n(x))_{n\geq 0}$ denote its sequence of $h$-polynomials. The coefficients of the polynomial $h_n(x)$ can be expressed as*

$$[x^k]\, h_n(x) \;=\; \sum_{S \in \binom{[0,n-1]}{k}} Q \begin{bmatrix} S \cup \{n\} \\ \{0\} \cup (S+1) \end{bmatrix} \tag{3.6}$$

*where the sum is over all subsets $S$ of size $k$ of the set $\{0, \ldots, n-1\}$ and the summand is the minor of the matrix $Q$ corresponding to rows $S \cup \{n\}$ and columns $\{0\} \cup (S+1)$.*

Proof. We prove this using induction. For $n = 0$, $h_0(x) = q_{00}$ which is exactly the minor of $Q$ corresponding to $S = \varnothing, n = 0$. Let us now assume that the formula holds for $n \geq 0$ and we will establish it for $n + 1$.

We rewrite (3.4b) as $\widehat{A}_{n+1}(x)(1 - q_{n+1,n+2}x) = q_{n+1,0} + x \sum_{i=0}^{n} q_{n+1,i+1}\, \widehat{A}_i(x)$ and multiply $\prod_{i=0}^{n}(1 - q_{i,i+1}x)$ on both sides to obtain

$$\begin{aligned}
h_{n+1}(x) \;&=\; q_{n+1,0} \prod_{i=0}^{n}(1 - q_{i,i+1}x) \;+\; x \sum_{i=0}^{n} q_{n+1,i+1}\, h_i(x) \prod_{j=i+1}^{n} (1 - q_{i,i+1}x) \\
&=\; (1 - q_{n,n+1}x) \left( q_{n+1,0} \prod_{i=0}^{n-1}(1 - q_{i,i+1}x) \;+\; x \sum_{i=0}^{n-1} q_{n+1,i+1}\, h_i(x) \prod_{j=i+1}^{n} (1 - q_{i,i+1}x) \right) \\
&\qquad + x q_{n+1,n+1} h_n(x) \\
&=\; h_n(x)\big|_{q_{n,i} \mapsto q_{n+1,i}}\, (1 - q_{n,n+1}x) \;+\; x q_{n+1,n+1} h_n(x)\;.
\end{aligned} \tag{3.7}$$

Now we extract the coefficient of $x^k$ on both sides of this equation to obtain

$$\begin{aligned}
[x^k]\, h_{n+1}(x) \;&=\; \sum_{S \in \binom{[0,n-1]}{k}} Q \begin{bmatrix} S \cup \{n+1\} \\ \{0\} \cup (S+1) \end{bmatrix} - \sum_{S \in \binom{[0,n-1]}{k-1}} q_{n,n+1} Q \begin{bmatrix} S \cup \{n+1\} \\ \{0\} \cup (S+1) \end{bmatrix} \\
&\qquad + \sum_{S \in \binom{[0,n-1]}{k-1}} q_{n+1,n+1} Q \begin{bmatrix} S \cup \{n\} \\ \{0\} \cup (S+1) \end{bmatrix} \\
&=\; \sum_{S \in \binom{[0,n-1]}{k}} Q \begin{bmatrix} S \cup \{n+1\} \\ \{0\} \cup (S+1) \end{bmatrix} + \sum_{S \in \binom{[0,n-1]}{k-1}} Q \begin{bmatrix} S \cup \{n, n+1\} \\ \{0, n+1\} \cup (S+1) \end{bmatrix} \\
&=\; \sum_{S \in \binom{[0,n]}{k}} Q \begin{bmatrix} S \cup \{n, n+1\} \\ \{0, n+1\} \cup (S+1) \end{bmatrix}
\end{aligned} \tag{3.8}$$

This finishes the proof. □

An immediate consequence of Lemma 3.7 is the following observation.

**Proposition 3.8.** *Let $Q = (q_{n,k})_{n,k\geq 0}$ be a unit lower-Hessenberg left production matrix and let $(h_n(x))_{n\geq 0}$ denote its sequence of $h$-polynomials. If $Q$ is totally positive, then for all $n \in \mathbb{N}$, the coefficients of $h_n(x)$ are positive in the underlying partially ordered commutative ring.*

Let the left production matrix $Q = \widehat{R}$ as in the setting of Lemma 3.3. Furthermore, assume that $R$ is a unit-lower triangular matrix, making $Q$ unit-lower Hessenberg, and also that the entries are over the reals. In this case, Bränden and Saud Maia Leite showed that if the matrix $R$ is totally positive then the $h$-polynomials are negative-real-rooted [3, Proof of Theorem 4.3]. In particular, this implies that the coefficients of $h_n$ are non-negative. Motivated by their result we leave the following open problem:

**Problem 3.9.** *Let $Q = (q_{nk})_{n,k\geq 0}$ be a lower-Hessenberg left production matrix and let $(h_n(x))_{n\geq 0}$ be its corresponding sequence of $h$-polynomials. Then find necessary and sufficient conditions on the matrix $Q$ such that for each $n \geq 0$, the coefficient sequence $\big([x^k]h_n(x)\big)_{k\geq 0}$ is Toeplitz totally positive.*

At this moment, we are unable to address this question.

### 3.2.2 Chain polynomials

For a given unit-lower triangular matrix $R$, Bränden and Saud Maia Leite [3] recently introduced the notion of *chain polynomials* which we now recall.

**Definition 3.10.** *For a unit-lower triangular matrix $R = (r_{n,k})_{n,k\geq 0}$, the chain polynomials associated to $R$ is the sequence $(p_n(x))_{n\geq 0}$ defined by $p_0(x) = 1$, and for $n \geq 0$,*

$$p_{n+1}(x) \stackrel{\text{def}}{=} x \sum_{i=0}^{n} r_{n+1,i}\, p_i(x) \; . \tag{3.9}$$

*The chain coefficient matrix $C$ is the matrix whose $(n,k)$-th entry is the coefficient $[x^k]p_n(x)$.*

Our first observation here is that this construction of Bränden and Leite can be rewritten in our theory of left-production matrices as follows:

**Proposition 3.11** (Chain polynomials from left output matrix)**.** *Let $\widetilde{R}$ denote the matrix $\widetilde{R} = (\mathbf{e}_{00} + (R - I)\,\Delta)$ . (Here $\widetilde{R}$ is lower triangular, and $(n,k)$-th entry for $n \geq 1$ and $1 \leq k \leq n$ is $r_{n,k-1}$.) Considering the matrix $\widetilde{R}$ as a left production matrix gives us the chain coefficient matrix $C$ as its left output matrix.*

Proof. Let $\widetilde{A}$ denote the left output matrix of $\widetilde{R}$ and let $\widetilde{A}_n(x)$ denote the $n$-th row-generating polynomial of $\widetilde{A}$. We rewrite the recurrence (3.4) in this setting to obtain

$$\widetilde{A}_0(x) \;=\; 1 \tag{3.10a}$$

$$\widetilde{A}_{n+1}(x) \;=\; x \sum_{i=0}^{n} r_{n+1,i}\widetilde{A}_i(x) \; , \tag{3.10b}$$

which is the same recurrence as in 3.9 and thus, we conclude $C = \widetilde{A}$. $\square$

Before proceeding we recall one of the main results in [3].

**Theorem 3.12** ( [3, Theorem 3.7]). *Let $R = (r_{n,k})_{n,k\geq 0}$ be a unit lower-triangular matrix with real entries. If $R$ is totally positive, then for each $n \in \mathbb{N}$, the zeros of the chain polynomial $p_n(x)$ are real and located in the interval $[-1, 0]$.*

Their theorem is extremely surprising from our point of view! Notice that according to Theorem 2.3(c) and Proposition 3.11, if the matrix $\widetilde{R}$ is totally positive, which is equivalent to the total positivity of $R - I$, it would imply that the chain polynomials $p_n(x)$ are real-rooted with zeros in the interval $(-\infty, 0]$. However, here the hypothesis is that $R$, and not $R - I$, is totally positive. Indeed, for a unit lower triangular matrix of real numbers $A$, the total positivities of the matrix $A$ and $A - I$ are not related to each other. For example, let $A$ be the following $5 \times 5$ matrix

$$A = \begin{bmatrix} 1 & & & & \\ 1 & 1 & & & \\ 1 & 2 & 1 & & \\ 1 & 3 & 3 & 1 & \\ 0 & 1 & 2 & 1 & 1 \end{bmatrix}. \tag{3.11}$$

Even though the matrix $A$ is totally positive, the matrix $A - I$ is not. In fact, the only real number $a$ such that $A - aI$ is totally positive is when $a = 0$.

**NOW WHAT????????** It seems the following relation between h-polynomials and chain polynomials holds for $R$ unit lower-triangular: $p_n(x) = (1 + x)^n h_n(x/(1 + x))$. Thus, $p_n$ is always $TP_1$ whenever $R$ is TP.

## 3.3 Toeplitz left production matrices

We first establish the following lemma.

**Lemma 3.13.** *Consider the sequence $(f_n)_{n\geq 0}$ with the ordinary generating function $f(t) = \sum_{n=0}^{\infty} f_n t^n$, and also let $F = (f_{n-k})_{n,k\leq 0}$ be its lower triangular Toeplitz matrix. Consider the following formal power series*

$$\frac{f(t)}{1 - xtf(t)} = \sum_{n=0}^{\infty} \widehat{a}_n(x) t^n \,, \tag{3.12a}$$

$$\frac{1}{1 - xtf(t)} = \sum_{n=0}^{\infty} \widehat{b}_n(x) t^n \,. \tag{3.12b}$$

*Here the coefficients $\widehat{a}_n(x), \widehat{b}_n(x)$ of the two series on the right hand side are polynomials in the variable $x$, respectively. Then we have*

(a) *The polynomials $\widehat{a}_n(x)$ are the row-generating polynomials of the left output matrix of $F$ considered as a left production matrix.*

(b) *Let $\widehat{F}$ be the matrix*

$$\widehat{F} = \left[\begin{array}{c|c} 1 & \\ \hline & F \end{array}\right] . \tag{3.13}$$

*The polynomials $\widehat{b}_n(x)$ are the row-generating polynomials of the left output matrix of $\widehat{F}$ considered as a left production matrix.*

Proof. Let $A$ and $\widehat{A}$ be the left-output matrices of $F$ and $\widehat{F}$, respectively. As $F, \widehat{F}$ are lower triangular so are $A, \widehat{A}$, respectively. Also, let $A_n(x), \widehat{A}_n(x)$ denote the row-generating polynomial of the $n$-th row of $A, \widehat{A}$, respectively. We rewrite the recurrence (3.4) in this setting to obtain for $F$

$$A_0(x) = f_0 \tag{3.14a}$$

$$A_{n+1}(x) = f_{n+1} + x \sum_{i=0}^{n} f_{n-i} A_i(x) , \tag{3.14b}$$

and for $\widehat{F}$, we obtain

$$\widehat{A}_0(x) = 1 \tag{3.15a}$$

$$\widehat{A}_{n+1}(x) = x \sum_{i=0}^{n} f_{n-i} \widehat{A}_i(x) . \tag{3.15b}$$

Rewriting (3.16a) as $f(t) = (1 - xtf(t))(\sum_{n=0}^{\infty} \widehat{a}_n(x)t^n)$, and (3.16b) as $1 = (1 - xtf(t))(\sum_{n=0}^{\infty} \widehat{b}_n(x)t^n)$, and comparing the coefficients of $t^0$ and $t^{n+1}$ from both sides in each equation establishes $\widehat{a}_n(x) = A_n(x)$ and $\widehat{b}_n(x) = \widehat{A}_n(x)$ as they satisfy the same recurrences. □

As a consequence of Lemma 3.13, we have the following result that is motivated by [3, Theorem 4.4].

**Theorem 3.14.** *Consider the sequence $(f_n)_{n\geq 0}$ with the ordinary generating function $f(t) = \sum_{n=0}^{\infty} f_n t^n$, and also let $F = (f_{n-k})_{n,k\leq 0}$ be its lower triangular Toeplitz matrix. Consider the following formal power series*

$$\frac{f(t)}{1 - xtf(t)} = \sum_{n=0}^{\infty} \widehat{a}_n(x) t^n , \tag{3.16a}$$

$$\frac{1}{1 - xtf(t)} = \sum_{n=0}^{\infty} \widehat{b}_n(x) t^n . \tag{3.16b}$$

*If $(f_n)_{n\geq 0}$ is Toeplitz totally positive of order $r$, then we have*

(a) *For each $n$ the coefficient sequence $\left([x^i]\,\widehat{a}_n(x)\right)_{i\geq 0}$ is Toeplitz totally positive of order $r$.*

(b) *For each $n$ the coefficient sequence $\left([x^i]\,\widehat{b}_n(x)\right)_{i\geq 0}$ is Toeplitz totally positive of order $r$.*

Proof. The proof follows by combining Lemma 3.13 and Theorem 2.3(c). □

# 4 Application: total positivity of the matrices $\boldsymbol{T}(0, c, d, e)$ and $\boldsymbol{T}(a, c, 0, e)$ in [6]

In [6], we considered a lower triangular matrix $\boldsymbol{T} = \big(T(n,k)\big)_{n,k\geq 0}$ whose entries are defined by the following linear recurrence,

$$T(n,k) \;=\; [a(n{-}k){+}c]\,T(n{-}1,k{-}1) + (dk{+}e)\,T(n{-}1,k) + [f(n{-}2){+}g]\,T(n{-}2,k{-}1) \tag{4.1}$$

for $n \geq 1$, with initial conditions $T(0,k) = \delta_{k0}$ and $T(-1,k) = 0$. We conjectured [6, Conjecture 1.4] that this matrix is totally positive in all six indeterminates $a, c, d, e, f, g$. This conjecture generalises Brenti's Eulerian triangle conjecture [5] which can be seen by substituting $a = c = d = e = 1$ and $f = g = 0$ in the matrix $\boldsymbol{T}$.

Our main result in [6, Theorems 1.5/3.2] was proving the total positivity of the matrix $\boldsymbol{T}(a, c, 0, e)$ which is obtained by substituting $d = f = g = 0$ in the matrix $\boldsymbol{T}$; we proved this by constructing an explicit *planar network* and we showed that its *path matrix* is the the matrix $\boldsymbol{T}(a, c, 0, e)$. We will now apply the method of left-production matrices and give a new proof of this result. In [7, sec. 4], The left-production-matrix method was used to prove the total positivity of the matrix $\boldsymbol{T}(1, 1, 0, 1)$, (the reversed Stirling subset matrix).

We first consider the matrix $\boldsymbol{T}(0, c, d, e)$; this the matrix $\boldsymbol{T}$ along with the substitution $a = f = g = 0$. In this special case, the recurrence (4.1) becomes purely $k$-dependent and is therefore coefficientwise totally positive [4]. Also, notice that $\overleftarrow{\boldsymbol{T}(0, c, d, e)} = \boldsymbol{T}(d, e, c, 0)$. We will now construct the left-production matrix of the matrix $\boldsymbol{T}(0, c, d, e)$ and prove that it is totally positive; we then use Theorem 2.3 to infer the total positivity of both matrices $\boldsymbol{T}(0, c, d, e)$ and $\boldsymbol{T}(a, c, e, 0)$ (after a relabeling of parameters).

We first introduce some notation. Let $T_\xi$ be the ***lower-triangular Toeplitz matrix of powers***, with matrix elements

$$(T_\xi)_{nk} \;=\; \begin{cases} \xi^{n-k} & \text{if } n \geq k \\ 0 & \text{otherwise} \end{cases} . \tag{4.2}$$

Next, we consider the ***weighted binomial matrix*** $B_{x,y} = \big((B_{x,y})_{nk}\big)_{n,k\geq 0}$ defined by

$$(B_{x,y})_{nk} \;=\; \binom{n}{k}\, x^{n-k}\, y^k \,. \tag{4.3}$$

We also write $B_x$ as a shorthand for $B_{x,1}$. It is well known that both matrices $T_\xi$ and $B_{x,y}$ are coefficientwise totally positive in the variables $\xi$ and $x, y$, respectively.

We are now ready to state the left-production matrix of $\boldsymbol{T}(0, c, d, e)$.

**Proposition 4.1** (Left-production matrix for $\boldsymbol{T}(0, c, d, e)$)**.** *The matrix*

$$Q \;\stackrel{\text{def}}{=}\; T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,B_d \end{array}\right] \tag{4.4}$$

*satisfies*

$$\boldsymbol{T}(0,c,d,e) \;=\; Q \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & \boldsymbol{T}(0,c,d,e) \end{array}\right] . \tag{4.5}$$

*That is, the left-production matrix $Q$ defined in (4.4) has $\boldsymbol{T}(0,c,d,e)$ as its left-output matrix.*

We will provide two proofs of Proposition 4.1 in Section 4.1.

From the total positivity of the matrices $T_e$ and $B_d$, we can immediately conclude:

**Proposition 4.2.** *The matrix $Q$ defined in (4.4) is coefficientwise totally positive in the indeterminates $c,d,e$.*

Finally, we use Theorem 2.3 to obtain the following theorem:

**Theorem 4.3.** *The matrices $\boldsymbol{T}(0,c,d,e)$ and $\boldsymbol{T}(d,e,c,0)$ are totally positive in the indeterminates $a,c,d,e$. Furthermore, for each row $n \ge 1$, the infinite upper triangular Toeplitz matrix $T_\infty((T(n,k))_{k\ge 0}) = (T(n,i-j))_{i,j\ge 0}$ is coefficientwise Toeplitz-totally positive in the indeterminates $a,c,e$.*

**Remark.** The coefficientwise Toeplitz-total positivity of the sequence $T_\infty((T(n,k))_{k\ge 0})$ is a vast generalization of the real-rootedness of the Stirling subset polynomials. ■

## 4.1 Proofs of Proposition 4.1

In this section, we will now provide two proofs of Proposition 4.1, one algebraic and one bijective.

We require Lemmas 4.4–4.6 for our first proof.

**Lemma 4.4** (Factorization of $B_{x,y}$ in terms of $T_x$)**.** *We have*

$$B_{x,y} \;=\; T_x \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & y\,B_{x,y} \end{array}\right] \tag{4.6}$$

*and hence*

$$B_{x,y} \;=\; T_x \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & y\,T_x \end{array}\right] \left[\begin{array}{c|c} I_2 & \mathbf{0} \\ \hline \mathbf{0} & y\,T_x \end{array}\right] \left[\begin{array}{c|c} I_3 & \mathbf{0} \\ \hline \mathbf{0} & y\,T_x \end{array}\right] \cdots \tag{4.7}$$

*where $I_k$ is the $k \times k$ identity matrix.*

Proof. Let $L_{-x}$ be the matrix with 1 on the diagonal, $-x$ on the subdiagonal, and 0 elsewhere; we have $T_x = (L_{-x})^{-1}$. Then

$$(L_{-x}B_{x,y})_{nk} \;=\; \binom{n}{k} x^{n-k}\, y^k \;-\; x \binom{n-1}{k} x^{n-1-k}\, y^k \tag{4.8a}$$

$$\;=\; \binom{n-1}{k-1} x^{n-k}\, y^k \;, \tag{4.8b}$$

which is precisely the $(n,k)$ matrix element of $\left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & y\, B_{x,y} \end{array}\right]$. This proves (4.6). Then (4.7) follows by iteration (note that each matrix element stabilizes after a finite number of factors). $\square$

**Lemma 4.5** (Binomial transform)**.** *We have*

$$B_{\xi,\eta}\, \boldsymbol{T}(0,c,d,e) \;=\; \boldsymbol{T}(0,\, \eta c,\, \eta d,\, \eta e + \xi)\;. \tag{4.9}$$

*In particular, when $\xi = -e$ and $\eta = 1$, we have*

$$\boldsymbol{T}(0,c,d,e) \;=\; B_e\, \boldsymbol{T}(0,c,d,0)\;. \tag{4.10}$$

We remark that Lemma 4.5 is (when normalized to $\eta = 1$) a special case of [24, Corollary A.11]. For completeness we include the proof:

Proof of Lemma 4.5. Writing $B = B_{\xi,\eta}$ for conciseness, the matrix $\boldsymbol{M} = B_{\xi,\eta}\, \boldsymbol{T}(0,c,d,e)$ satisfies

$$\begin{aligned}
M(n,k) \;&=\; \sum_{j=0}^{n} B(n,j)\, T(j,k) && (4.11a)\\
&=\; \sum_{j=0}^{n} \big[\xi\, B(n-1,j) + \eta B(n-1,j-1)\big]\, T(j,k) && (4.11b)\\
&=\; \xi\, M(n-1,k) + \eta \sum_{j=1}^{n} B(n-1,j-1) \times \Big[c\, T(j-1,k-1) \\
&\qquad\qquad + (dk+e)\, T(j-1,k)\Big] && (4.11c)\\
&=\; c\eta\, M(n-1,k-1) \;+\; \big[(dk+e)\eta + \xi\big]\, M(n-1,k)\;, && (4.11d)
\end{aligned}$$

which implies (4.10). $\square$

**Lemma 4.6** (Pulling out a Toeplitz matrix of powers)**.** *We have*

$$\boldsymbol{T}(0,c,d,e) \;=\; T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\, \boldsymbol{T}(0,c,d,d+e) \end{array}\right]. \tag{4.12}$$

The proof of Lemma 4.6 will be based on Lemma 4.5 combined with the identity (4.7) that relates the $e$-binomial matrix to the $e$-Toeplitz matrix of powers.

Proof of Lemma 4.6. As special cases of (4.9), we have

$$B_e\, \boldsymbol{T}(0,c,d,0) \;=\; \boldsymbol{T}(0,c,d,e) \tag{4.13}$$

$$B_e\, \boldsymbol{T}(0,c,d,d) \;=\; \boldsymbol{T}(0,c,d,d+e) \tag{4.14}$$

We further claim that the following identity is also true:

$$\boldsymbol{T}(0,c,d,0) \;=\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,\boldsymbol{T}(0,c,d,d) \end{array}\right] . \tag{4.15}$$

It is clear that the matrix $\boldsymbol{T}(0,c,d,0)$ can be written as $\boldsymbol{T}(0,c,d,0) = \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & \boldsymbol{U} \end{array}\right]$ where $U(n,k) = T(n+1,k+1;\, 0,c,d,0)$. Inserting the recurrence (4.1) and using $U(-1,-1) = T(0,0) = 1$, we obtain

$$U(n,k) \;=\; c\,U(n-1,k-1) \;+\; (dk+d)\,U(n-1,k) \;+\; c\delta_{n0}\delta_{k0} \,. \tag{4.16}$$

This shows that $U(n,k) = c\,T(n,k;\, 0,c,d,d)$ which implies our claim.

Finally, obtain the desired identity as follows

$$\begin{aligned} &\boldsymbol{T}(0,c,d,e) \\ &\quad = B_e\,\boldsymbol{T}(0,c,d,0) \qquad \text{by (4.13)} && \text{(4.17a)} \\ &\quad = T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & B_e \end{array}\right] \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,\boldsymbol{T}(0,c,d,d) \end{array}\right] \qquad \text{by (4.7) and (4.15)} && \text{(4.17b)} \\ &\quad = T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,\boldsymbol{T}(0,c,d,d+e) \end{array}\right] \qquad \text{by (4.14)}\,. && \text{(4.17c)} \end{aligned}$$

□

We are now ready to prove Proposition 4.1.

Algebraic Proof of Proposition 4.1. Using the identities in Lemmas 4.6 and 4.5, we have

$$\begin{aligned} \boldsymbol{T}(0,c,d,e) \;&=\; T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,\boldsymbol{T}(0,c,d,d+e) \end{array}\right] && \text{(4.18a)} \\ &=\; T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,B_d\,\boldsymbol{T}(0,c,d,e) \end{array}\right] && \text{(4.18b)} \\ &=\; T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,B_d \end{array}\right] \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & \boldsymbol{T}(0,c,d,e) \end{array}\right] . && \text{(4.18c)} \end{aligned}$$

□

We now recall [6, Lemma 2.2(i)] which we require for our second proof.

**Lemma 4.7** ( [6, Lemma 2.2(i)]). *The matrix $\boldsymbol{T} = \boldsymbol{T}(0, c, d, e)$ satisfies the recurrence*

$$T(n,k) \;=\; e\,T(n-1,k) \,+\, \sum_{m=0}^{n-1} \binom{n-1}{m} d^m c\,T(n-1-m,k-1) \tag{4.19}$$

*for $n \geq 1$, where $T(n,k) \stackrel{\text{def}}{=} 0$ if $n < 0$ or $k < 0$.*

Second Proof of Proposition 4.1. We rewrite the recurrence (4.19) as

$$T(n,k) \,-\, e\,T(n-1,k) \;=\; c \sum_{m=0}^{n-1} \binom{n-1}{n-1-m} d^m\, T(n-1-m,k-1)\;, \tag{4.20}$$

which in matrix form reads

$$L_{-e}\, \boldsymbol{T}(0,c,d,e) \;=\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,B_d \end{array}\right] \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & \boldsymbol{T}(0,c,d,e) \end{array}\right] . \tag{4.21}$$

Left-multiplying both sides by $(L_{-e})^{-1} = T_e$ yields (4.5). □

## 4.2 A conjectural generalisation of the real-rootedness of the Eulerian polynomials

For $n \geq 0$, let $T_n(x) = \sum_{k=0}^{n} T(n,k) x^k$ denote the generating polynomial of the $n$-th row of the matrix $\boldsymbol{T}$ defined by the linear recurrence (4.1). We have the following recurrence:

**Proposition 4.8.** *The polynomials $T_n(x) = \sum_{k=0}^{n} T(n,k) x^k$ where the coefficients are given by* (4.1)*, satisfy the following recurrence relation:*

$$T_n(x) \;=\; \Big(\big(a(n-1)+c\big)\,x + e\Big)\,T_{n-1}(x) + x\,(d-ax)\,T'_{n-1}(x) + \big(f(n-2)+g\big)\,x\,T_{n-2}(x)\;. \tag{4.22}$$

The proof of Proposition 4.8 is straightforward and we omit the details. However, this recurrence gives us the following more interesting consequence.

**Proposition 4.9.** *Let $a, c, d, e, f, g$ be non-negative real-numbers. Then for all $n \geq 0$, the polynomial $T_n(x)$ is real-rooted.*

PROOF. Define polynomials $p(x), q(x)$ by $p(x) = x(d-ax)$ and $q_n(x) = (f(n-2)+g)x$. For every $x \le 0$, we always have $p(x) \le 0$ and $q(x) \le 0$. The proposition then follows from [19, Corollary 2.4]. $\square$

Proposition 4.9 naturally suggests the following upgrade:

**Conjecture 4.10.** *Let $a, c, d, e, f, g$ be indeterminates. For every fixed $n \ge 0$, the Toeplitz matrix $T_\infty((T(n,k))_{k\ge 0})$ is coefficientwise totally positive in all six indeterminates $a, c, d, e, f, g$.*

Notice that the two cases $a = f = g = 0$ and $d = f = g = 0$ in Conjecture 4.10 are covered by Theorem 4.3.

# 5 Left production matrices for ordinary and Exponential Riordan arrays

**DEFINE RIORDAN ARRAYS!!!!!**

**Proposition 5.1.** *(a) Let $F = \sum_{n=0}^{\infty} f_n t^n$ and let $G = \sum_{n=0}^{\infty} g_n t^n$ be formal power series with $g_0 = 0$, and $f_0 \neq 0$ is invertible. The ordinary Riordan array $\mathcal{R}(F, G)$ has the following matrix as its left production matrix:*

$$Q \;=\; \mathcal{R}(F,t) \cdot \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}(G/(tF),t) \end{array}\right] \;=\; \widetilde{Q} \cdot \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}(G/t,t) \end{array}\right] \tag{5.1}$$

*where $\widetilde{Q}$ is the left production matrix of $\mathcal{R}(F,t)$.*

*(b) Let $F = \sum_{n=0}^{\infty} f_n t^n/n!$ and let $G = \sum_{n=0}^{\infty} g_n t^n/n!$ be formal power series with $g_0 = 0$, and $f_0 \neq 0$ is invertible. The exponential Riordan array $\mathcal{R}[F, G]$ has the following matrix as its left production matrix:*

$$Q \;=\; \mathcal{R}[F,t] \cdot \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}[G'/F,t] \end{array}\right] \;=\; \widetilde{Q} \cdot \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}[G',t] \end{array}\right] \tag{5.2}$$

*where $\widetilde{Q}$ is the left production matrix of $\mathcal{R}(F,t)$.*

PROOF. We only exhibit our computation for (a).

$$\begin{aligned} Q \;&=\; \mathcal{R}(F,G) \cdot \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}(F,G)^{-1} \end{array}\right] \\ &=\; \mathcal{R}(F,t) \cdot \mathcal{R}(1,G) \cdot \left[\begin{array}{c|c} 1 & \\ \hline & \\ & \mathcal{R}(1/F(\overline{G}),t) \end{array}\right] \end{aligned} \tag{5.3}$$

The first equality in (5.1) follows by writing $\mathcal{R}(1,G) = \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}(G/t,G) \end{array}\right]$ and then multiplying the three terms using the binary operation of the Riordan group. The second equality is the due to the fact that the matrices $\mathcal{R}(G/t, t)$ and $\mathcal{R}(1/F, t)$ commute and their matrix product is $\mathcal{R}(G/(tF), t)$.

The proof of (b) is similar and we omit the details. □

**Remarks.** Let us consider the situation when the exponential Riordan array $\mathcal{R}[F, G]$ belongs to the *derivative subgroup*, i.e., $F = G'$. In this case, as $\mathcal{R}[1, t]$ is the identity matrix, we get that the left production matrix of $\mathcal{R}[G', G]$ is simply $Q = \mathcal{R}[F, t]$; this was noticed in [7, Proof of Theorem 4.2].

Analogously, for ordinary Riordan arrays, when $\mathcal{R}(F, G)$ belongs to the *Bell subgroup*, i.e., $G = tF$, we get that the left production matrix of $\mathcal{R}(F, tF)$ is simply $Q = \mathcal{R}(F, t)$.

Notice that in both situations, the left production matrix is itself a (exponential or ordinary, resp.) Riordan array that belongs to the *Appell subgroup*. ■

**Example 5.2.** *For, $r, s \in \mathbb{N}$, consider the ordinary Riordan array $\mathcal{R}((1-t)^{-r}, t(1-t)^{-s})$; this matrix is the* Fuss–Catalan Riordan array $\mathcal{C}^{(1)}(r, s)$ *of Zhu in [36]. Here, $F(t) = (1-t)^{-s}$ is certainly a P.F. ordinary generating function which implies that the matrix $\mathcal{R}(F(t), t)$ is totally positive. Supposing $s \geq r$, we get that $G'(t)/(tF(t)) = (1-t)^{-(s-r)}$ is also a P.F. ordinary generating function which implies that the matrix $\mathcal{R}(G'(t)/(tF(t)), t)$ is totally positive. Thus, in this case the left production matrix $Q$ is totally positive.*

For ordinary *wide-sense* Riordan arrays, Zhu [36] recently discovered another expression for their left production matrix; extending a result due to Mao, Mu and Wang [20]. We will now write this out. Let $F(t) = \sum_{n=0}^{\infty} f_n t^n$ and $G(t) = \sum_{n=0}^{\infty} g_n t^n$ be two ordinary generating functions. Let $\Gamma(f, g)$ be the following matrix

$$\Gamma(F, G) \stackrel{\text{def}}{=} \begin{bmatrix} f_0 & g_0 & & & \\ f_1 & g_1 & g_0 & & \\ f_2 & g_2 & g_1 & g_0 & \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \tag{5.4}$$

**CITE ALAN'S PAPER ON CHU–VANDERMONDE!!!!!!!!**

**Proposition 5.3.** *[36, eq. (3.4)] Let $F(t) = \sum_{n=0}^{\infty} f_n t^n$ and $G(t) = \sum_{n=0}^{\infty} g_n t^n$ be two ordinary generating functions. Consider the wide-sense Riordan array $\mathcal{R}(F, G) = \big([t^n]F(t)G(t)^k\big)_{n,k\geq 0}$; its left production matrix is the matrix $\Gamma(F, G)$.*

Proof. **WRITE SOME DETAILS!!!!!!!!** □

As the lower-triangular Toeplitz matrix $(g_{n-k})_{n,k\geq 0}$ is a submatrix of $\Gamma(F, G)$, from Proposition 5.3, we get that a necessary condition for the left production matrix of an ordinary Riordan array $\mathcal{R}(F, G)$ to be totally positive is for the sequence $(g_n)_{n\geq 0}$ to be Toeplitz totally positive.

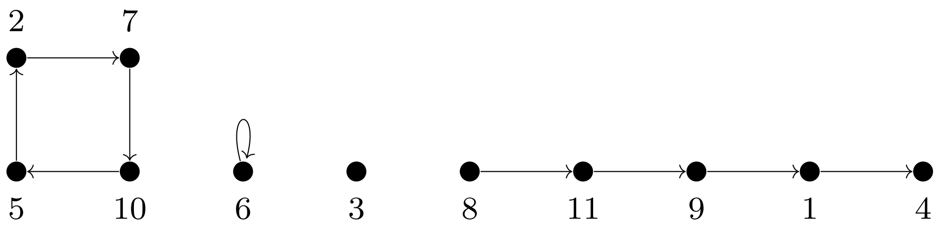


Figure 1: A Laguerre digraph on 11 vertices with 2 cycles (one of which is a loop) and 2 paths (one of which is an isolated vertex).

## 5.1 Application: Univariate Laguerre coefficient matrix

A ***Laguerre digraph*** is a directed graph in which each vertex has out-degree 0 or 1 and in-degree 0 or 1. It follows that each weakly connected component of a Laguerre digraph is either a directed path of some length $\ell \geq 0$ (where a path of length 0 is an isolated vertex) or else a directed cycle of some length $\ell \geq 1$ (where a cycle of length 1 is a loop): see Figure 1. Laguerre digraphs were introduced by Foata and Strehl [15] as Laguerre configurations; however the definition that we use here was first defined by Sokal in [25].

For each integer $n \geq 0$, let us write $\mathbf{LD}_n$ for the set of Laguerre digraphs on the vertex set $[n] \stackrel{\text{def}}{=} \{1, \ldots, n\}$; and for $n \geq k \geq 0$, let us write $\mathbf{LD}_{n,k}$ for the set of Laguerre digraphs on the vertex set $[n]$ with $k$ paths. For $G \in \mathbf{LD}_n$, we write $\mathrm{e}(G)$ [resp. $\mathrm{cyc}(G)$, $\mathrm{pa}(G)$] for the number of edges (resp. cycles, paths) in $G$, and observe that $\mathrm{e}(G) = n - \mathrm{pa}(G)$.

In this paper, the ***monic unsigned Laguerre polynomials*** are defined by

$$\begin{aligned} \mathcal{L}_n^{(\lambda)}(x) \;&\stackrel{\text{def}}{=}\; \sum_{k=0}^{n} \frac{n!}{k!} \binom{n-1+\lambda}{n-k} x^k \;=\; \sum_{k=0}^{n} \binom{n}{k} (n-1+\lambda)^{\underline{n-k}} x^k \\ &=\; \lambda^{\overline{n}} \, {}_1F_1\!\left( \begin{array}{c} -n \\ \lambda \end{array} \middle|\, -x \right) \end{aligned} \tag{5.5}$$

[where $\rho^{\underline{n}} \stackrel{\text{def}}{=} \rho(\rho-1)\cdots(\rho-n+1)$ and $\rho^{\overline{n}} \stackrel{\text{def}}{=} \rho(\rho+1)\cdots(\rho+n-1)$]. Foata and Strehl [15] provided a beautiful combinatorial interpretation for the polynomials $\mathcal{L}_n^{(\lambda)}(x)$:

$$\mathcal{L}_n^{(\lambda)}(x) \;=\; \sum_{G \in \mathbf{LD}_n} x^{\mathrm{pa}(G)} \, \lambda^{\mathrm{cyc}(G)} \,. \tag{5.6}$$

In this section, we will work with the ***Laguerre coefficient matrix***: it is a unit-lower-triangular matrix $\mathsf{L}^{(\lambda)}$ with entries

$$\mathsf{L}_{n,k}^{(\lambda)} \;=\; \frac{n!}{k!} \binom{n-1+\lambda}{n-k} \;=\; \binom{n}{k} (n-1+\lambda)^{\underline{n-k}} \;=\; \sum_{G \in \mathbf{LD}_{n,k}} \lambda^{\mathrm{cyc}(G)} \,. \tag{5.7}$$

For the Laguerre polynomials $\mathcal{L}_n^{(\lambda)}(x)$, it is well known that they have the following exponential generating function (see [2], for instance)

$$\sum_{n=0}^{\infty} \mathcal{L}_n^{(\lambda)}(x) \frac{t^n}{n!} \;=\; (1-t)^{-\lambda} \exp\left(\frac{xt}{1-t}\right) \,. \tag{5.8}$$

A consequence of this formula is that the Laguerre coefficient matrix forms an exponential Riordan array; this was observed in [9, 34, 35]. We state this fact in the following proposition.

**Proposition 5.4** ( [9, 34, 35])**.** *The Laguerre coefficient matrix $\mathsf{L}^{(\lambda)}$ is equal to the exponential Riordan array $\mathcal{L}\big[(1-t)^{-\lambda}, t/(1-t)\big]$.*

In the rest of this section, we will compute the left production matrix of $\mathsf{L}^{(\lambda)}$ and prove that it is totally positive.

**Lemma 5.5.** *Let $T = (t_{n,k})_{n,k\ge 0}$ be the lower-triangular tridiagonal matrix with entries*

$$t_{k,k} = 1\,, \qquad t_{k+1,k} = -\lambda - 2k\,, \qquad t_{k+2,k} = (k+1)(k+\lambda)\,. \tag{5.9}$$

*The matrix inverse $T^{-1}$ is the left production matrix of the Laguerre coefficient matrix $\mathsf{L}^{(\lambda)}$.*

Proof. **WRITE OUT DETAILS!!!!!!!!** □

Finally, we show that the left production matrix is totally positive.

**Proposition 5.6.** *Let $T$ be the matrix defined in Lemma 5.5. The matrix $Q = T^{-1}$, which is the left production matrix of the Laguerre coefficient matrix $\mathsf{L}^{(\lambda)}$, is totally positive.*

Proof. Let $D$ be the diagonal matrix with diagonal entries $\big((-1)^n\big)_{n\ge 0}$. We first show that the matrix $\widetilde{T} = DTD$ is totally positive; this matrix is tridiagonal with entries $\widetilde{T} = (\widetilde{t}_{nk})_{n,k\ge 0}$ given by

$$\widetilde{t}_{k,k} = 1\,, \qquad \widetilde{t}_{k+1,k} = \lambda + 2k\,, \qquad \widetilde{t}_{k+2,k} = (k+1)(k+\lambda)\,. \tag{5.10}$$

Let $L_1, L_2$ be unit lower bidiagonal matrices with subdiagonal entries given by sequences $(i)_{i\ge 0}$ and $(\lambda + i)_{i\ge 0}$, respectively. It is easy to check that the matrix $\widetilde{T}$ has the following bidiagonal factorisation

$$\widetilde{T} \;=\; L_1\, L_2\,, \tag{5.11}$$

which implies that $\widetilde{T}$ is totally positive. As $Q$ is the unsigned inverse of $\widetilde{T}$, it follows that $Q$ is also totally positive. □

**Remark.** We remark that when $\lambda = 1$, the left production matrix $Q$ here is the triangle [21, A165675] whose row sums are [21, A093345]. ■

At this moment, we do not have a complete bijective understanding of Lemma 5.5. Therefore, we ask the following open problem.

**Problem 5.7.** *Find a combinatorial interpretation of the left production matrix $Q$. Using this combinatorial interpretation, construct a bijective proof of the fact that $Q$ is the left production matrix of $\mathsf{L}^{(\lambda)}$.*

Finally, we conclude this section by combining Lemma 5.5 and Proposition 5.6 along with Theorem 2.3 to obtain the following result.

**Theorem 5.8.** *The following statements hold:*

*(a) The matrix $\mathsf{L}^{(\lambda)}$ is totally positive with respect to the variable $\lambda$.*

*(b) The reversal $\overleftarrow{\mathsf{L}^{(\lambda)}}$ is totally positive with respect to the variable $\lambda$.*

*(c) For any $n$, the infinite lower triangular Toeplitz matrix of the sequence $(\mathsf{L}_{n,k})_{k\geq 0} = \Big([x^k]\mathcal{L}_n^{(\lambda)}\Big)_{k\geq 0}$ is totally positive.*

**Remarks.** 1. We have obtained a new proof of the fact that the Laguerre coefficient matrix is totally positive; this was previously proved in [9, 34, 35].

2. It is well known that the Laguerre polynomials are real rooted whenever the parameter $\lambda \geq 0$ (classically, $\lambda$ is replaced with $1+\alpha$ and the condition is $\alpha \geq -1$). The third statement is a coefficientwise generalization of this classical fact.

3. As far as we know, our observation of statements (b) and (c) in Theorem 5.8 are new.

4. In [9], we work with various multivariate generalizations of the matrix $\mathsf{L}^{(\lambda)}$ counting several statistics on Laguerre digraphs. We showed in vast generality that the row-generating polynomials of these matrices are coefficientwise Hankel totally positive by using the theory of right production matrices for exponential generating functions. A natural question is if one can further generalize our total positivity result in Proposition 5.6 from a univariate setting to a multivariate setting. We experimented in the setting of the first multivariate coefficient matrix [9, sec. 1.3] which has three additional variables $v_0, v_-, v_+$ along with $\lambda$. We computed its left production matrix $Q(v_0, v_-, v_+)$ and tested for total positivity. The specialisation $Q(v, v, v)$ is a rescaling of the matrix $Q$ in Lemma 5.5. We observed that neither $Q(v_0, v_-, v_+)$, nor any of its specializations, except for the trivial specialisation, is totally positive. Thus, the hope for generalizing our result to a multivariate setting is grim. ■

# 6 Application: Linearly ordered phylogenetic trees

In [10, Final Remark in Section 4.3], the authors *phylogenetic trees* with linearly ordered children, and formulated [10, Conjectures 4.17]. We will prove [10, Conjectures 4.17(a)(b)] in this section.

We first recall the relevant combinatorial definitions. A ***phylogenetic tree*** is a leaf-labeled rooted tree (i.e. the leaves are labeled and the non-leaf vertices are unlabeled) in which all non-leaf vertices have at least two children. A ***linearly***

***ordered phylogenetic tree*** is a phylogenetic tree such that the children at each vertex are linearly ordered. Let $\mathcal{D}_{n,k}$ denote the set of all linearly ordered phylogenetic trees with $n+1$ leaves and $k$ internal (i.e. non-leaf) vertices; let $\widehat{\left\{ {n \atop k} \right\}}$ denote the cardinality of this set $\widehat{\left\{ {n \atop k} \right\}} := |\mathcal{D}_{n,k}|$. This number has a closed form expression as was observed in [10, Proposition 4.16, eq. (4.39)]:

$$|\mathcal{D}_{n,k}| \;=\; \widehat{\left\{ {n \atop k} \right\}} \;=\; \frac{(n+k)!}{k!}\binom{n-1}{k-1}\;; \tag{6.1}$$

these numbers match [21, A357367]. Let $d_n(x)$ denote the generating polynomials

$$d_n(x) \;\stackrel{\text{def}}{=}\; \sum_{k=0}^{n} \widehat{\left\{ {n \atop k} \right\}} x^k\;, \tag{6.2}$$

and let $D$ be the lower-triangular matrix $D = \left(\widehat{\left\{ {n \atop k} \right\}}\right)_{n,k\ge 0}$.

It was conjectured [10, Conjecture 4.17] that the matrices $D, \overleftarrow{D}$ are totally positive, the sequence $(d_n(x))_{n\ge 0}$ is coefficientwise Hankel totally positive, and it was proved that the polynomials $d_n(x)$ are negative-real-rooted [10, Theorem 4.18]. Here we will now prove that the matrices $D, \overleftarrow{D}$ are both totally positive and provide two proofs. Our first proof involves a matrix factorization for both matrices. For our second proof, we will compute the left production matrix of $D$ and show that it is totally positive; then by virtue of Theorem 2.3 we obtain that the matrices $D, \overleftarrow{D}$ are both totally positive and also a new proof of the real-rootedness of the polynomials $d_n(x)$.

**Theorem 6.1** (Total positivity for linearly ordered phylogenetic trees)**.** *The matrix $D = \left(\widehat{\left\{ {n \atop k} \right\}}\right)_{n,k\ge 0}$ and its reverse $\overleftarrow{D}$ are both totally positive.*

First Proof of Theorem 6.1. We first show that $D$ is totally positive. First let $M = \left(\binom{n+k}{n-k}\right)_{n,k\ge 0}$ be the Morgan–Voyce matrix which is known to be totally positive as follows: the matrix $M$ is the (ordinary) Riordan array

$$M \;=\; \mathcal{R}(1/(1-t), t/(1-t)^2) \tag{6.3}$$

and as the coefficients of $1/(1-t), t/(1-t)^2$ both form a Polya frequency sequence by using [8, Theorem 2.1], we get that $M$ is totally positive (see [8] for details).

Next, let $D_1, D_2$ be the diagonal matrices whose 00-entries are 1 and the other diagonal entries are given by the sequences $\big((n-1)!\big)_{n\ge 1}, \big(\frac{(2k)!}{k!(k-1)!}\big)_{k\ge 1}$. We now notice that

$$\widehat{\left\{ {n \atop k} \right\}} = (n-1)! \cdot \binom{n+k}{n-k} \cdot \frac{(2k)!}{k!(k-1)!}\;. \tag{6.4}$$

which gives us the matrix factorisation $D = D_1 \cdot M \cdot D_2$. As each of the constituent matrices in this product are totally positive, the matrix $D$ is totally positive.

Next, we show that $\overleftarrow{D}$ is totally positive. Here, let $N = \left(\frac{1}{k+1}\binom{n+1}{k}\binom{n}{k}\right)_{n,k\geq 0}$ be the matrix of Narayana numbers (of type A), and let $B = \left(\binom{n}{k}\right)_{n,k\geq 0}$ be the triangle of binomial coefficients. Finally, let $D_3$ be the diagonal matrix with diagonal entries given by the sequence $((n+2)!)_{n\geq 0}$. Here the matrix $\overleftarrow{D}$ has the following matrix factorisation:

$$\overleftarrow{D} \;=\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \multicolumn{2}{c}{D_3 \cdot N \cdot B} \end{array}\right] \;; \tag{6.5}$$

it is straightforward to check this factorisation and we omit the details. In [31], Wang and Yang showed that the Narayana triangle $N$ is totally positive. It is well known that the binomial matrix $B$ and also the matrix $D_3$ is clearly totally positive. All of this together implies that the matrix $\overleftarrow{D}$ is totally positive. $\square$

We will now state our second proof which is more relevant to the contents of this paper. More precisely, we will establish the following left production matrix:

**Proposition 6.2.** *(a) The matrix $D$ enumerating linearly ordered phylogenetic trees has the left production matrix $Q = (q_{nk})_{n,k\geq 0}$ whose entries $q_{nk}$ are given by*

$$q_{nk} \;=\; \begin{cases} 1 & \text{if } n = k = 0 \\ \dfrac{(n-1)!}{k!}\,(2k-1)\,(n+k)(n-k+1) & \text{if } n \geq k > 0 \\ 0 & \text{otherwise} \end{cases} \;. \tag{6.6}$$

*(b) The matrix $Q$ is totally positive.*

Proof of Proposition 6.2(a). We will show that the numbers $\widehat{\left\{ n \atop k \right\}}$ satisfy the following recurrence relation:

$$\widehat{\left\{ n+1 \atop k+1 \right\}} \;=\; \sum_{i=0}^{n} q_{n+1,i+1} \widehat{\left\{ i \atop k \right\}} \;=\; \sum_{i=k}^{n} \frac{n!}{(i+1)!}\,(2i{+}1)\,(n{+}i{+}2)(n{-}i{+}1)\,\widehat{\left\{ i \atop k \right\}}\;. \tag{6.7}$$

[2] Consider a tree $T \in \mathcal{D}_{n+1,k+1}$. **I AM STUCK NOW!!!! UGH!!!! I COULD USE WZ BUT I WILL GET NO COMBINATORIAL INSIGHT IF I DO THAT!!!! NEED BIJECTIVE PROOF OF THIS!!!!**
$\square$

[2]**Remark.** We remark that these numbers also satisfy the following recurrence:

$$\widehat{\left\{ n+1 \atop k+1 \right\}} \;=\; \sum_{i=k}^{n} \binom{n+k+1}{n+1-i} (n+2-i)! \widehat{\left\{ i \atop k \right\}}\,, \tag{6.8}$$

which is interesting because the coefficients multiplying the numbers $\widehat{\left\{ i \atop k \right\}}$ are different from those in (6.7).

Proof of Proposition 6.2(b). Let $D_1, D_2, D_3$ be the diagonal matrices whose diagonal entries are the three sequences $((n)!)_{n\geq 0}, (2n+2)_{n\geq 0}, ((2k+1)/(k+1)!)_{k\geq 0}$, respectively. Also, let $T$ denote the lower triangular matrix all of whose non-zero entries are 1. We claim that the left production matrix $Q$ has the following matrix factorisation:

$$Q \;=\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & D_1\, T\, D_2\, T\, D_3 \end{array}\right] . \tag{6.9}$$

This is straightforward, we compute the entries and then compare the resulting expression with (6.6). As each of the constituent matrices $D_1, D_2, D_3, T$ are totally positive, hence the matrix $Q$ is also totally positive. $\square$

Second Proof of Theorem 6.1. The theorem follows by combining Proposition 6.2 with Theorem 2.3(a),(b). $\square$

A further consequence of Proposition 6.2 combined with Theorem 2.3(c) is a new proof of the following fact that was previously observed in [10, Theorem 4.18]:

**Corollary 6.3.** *For every $n \geq 1$, the polynomials $d_n(x)$ are negative real-rooted.*

**Remark.** The matrix $\Delta \overleftarrow{D}$ is lower triangular with non-zero diagonal entries. We remark that its left production matrix is not even $\mathrm{TP}_1$. $\blacksquare$

We finish this section by stating another combinatorial interpretation for the numbers $\widehat{\left\{ {n \atop k} \right\}}$ that was observed in [32].

**Proposition 6.4.** *The number $\widehat{\left\{ {n \atop k} \right\}}$ counts the number of ordered set partitions of $n+k$ elements into $k$ blocks, where each block is of length greater than or equal to 2, such that the elements in each block have a total order among them.*

**WRITE THE PROOF WHICH IS VIA MULTIVARIATE WARD POLYNOMIALS!!!!**

# 7 No implications between the four positivity properties

Let $A = (a_{n,k})_{n,k\geq 0}$ be a lower-triangular matrix of real numbers; let $A_n(x) = \sum_{k=0}^{n} a_{n,k} x^k$ denote its $n$-th row-generating polynomial. We started this paper by considering the following four positivity questions:

(a) Is $A$ TP?

(b) Is $\overleftarrow{A}$ TP?

(c) Is $A_n(x)$ negative-real-rooted for every $n \in \mathbb{N}$?

(d) Is the sequence $(A_n(x))_{n\geq 0}$ coefficientwise Hankel TP?

It is natural to ask if any (or any strict subset) of these total positivities is true, whether or not it implies any of the other total positivities, i.e., if there are any dependencies between any of these four different questions. As the title of this section suggests, we will now show that there are no dependencies.

The set of all doubly infinite lower triangular matrices can be divided into 16 different classes depending on whether or not their answers to these four questions is true or false. We will use elements of $\{+,-\}^4$ to refer to each of these classes, $+$ for true and $-$ for false, in the order in which these questions have been stated. For example, $(-,+,+,-)$ will denote matrices $A$ which are not TP, but $\overleftarrow{A}$ is TP, the row-generating polynomials are all real rooted, and the sequence $(A_n(x))_{n\geq 0}$ is not coefficientwise Hankel TP.

In this section, we will produce an example for each of these 16 classes thereby showing that it is non-empty. Notice that taking the symmetry of $A$ and $\overleftarrow{A}$ into consideration, it suffices to work with only 12 of these.

$\underline{(+,+,+,+)}$: This class is clearly non-empty as we have already seen several examples of such matrices earlier in this paper. For example, the binomial triangle $\left(\binom{n}{k}\right)_{n,k\geq 0}$, the Stirling subset triangle $\left(\left[{n \atop k}\right]\right)_{n,k\geq 0}$, the Stirling cycle triangle $\left(\left\{{n \atop k}\right\}\right)_{n,k\geq 0}$.

$\underline{(-,-,-,-)}$: If any of the four questions is true, it implies that the matrix $A$ must be entrywise positive. Thus, a matrix that has at least one negative entry is an example from this class. Let us look at an entrywise positive example from this class. Consider any matrix $A$ with second, third and fourth row-generating polynomials given by $A_2(x) = 1+x$, $A_3(x) = 1+5x+5x^2+x^3$ and $A_4(x) = 5+x+x^2+x^3+5x^4$. Here, the coefficients of $A_4(x)$ are not even log-concave and thus, $A_4(x)$ cannot be negative-real-rooted. Both $A$ and $\overleftarrow{A}$ have $\begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}$ as a submatrix which is an obstruction to total positivity. Finally, notice that the expression $A_2(x)A_4(x) - (A_3(x))^2$ is not coefficientwise positive and thus, even log-convexity of the sequence $(A_n(x))_{n\geq 0}$ fails.

$\underline{(+,+,-,+)}$: Consider the matrix $A$ with entries $a_{nk} = 4^{n^2}$; it has row-generating polynomials

$$A_n(x) \;=\; 4^{n^2} \sum_{i=0}^{n} x^i \;. \tag{7.1}$$

Here, we have $A = \overleftarrow{A}$, and this matrix coincides upto a scaling of rows with the lower-triangular matrix of all ones. Thus, $A = \overleftarrow{A}$ is totally positive. Also, it is clear that these polynomials are not real-rooted for $n \geq 2$.

We will now show that the sequence $(A_n(x))_{n\geq 0}$ is coefficientwise Hankel totally positive. To do this we will show that the expression $A_{n-1}(x)A_{n+1}(x) - 4(A_n(x))^2$ is always coefficientwise positive for $n \geq 1$ and then apply the coefficientwise extension of the Katkova–Vishnyakova theorem **HOW TO CITE THIS?????????**. For the sake of brevity let $q_n = \sum_{i=0}^{n} x^i$. Then we have

$$\begin{aligned} A_{n-1}(x)A_{n+1}(x) &= 16\left(4^{n^2}\right)^2 (q_{n-1}q_{n+1}) \\ &= 16\left(4^{n^2}\right)^2 (q_n^2 \,-\, x^n q_n \,+\, q_{n-1}x^{n+1}) \\ &= 4(A_n(x))^2 \,+\, 4\left(4^{n^2}\right)^2 \left(3q_n^2 \,-\, 4(x^n q_n \,-\, q_{n-1}x^{n+1})\right) \end{aligned} \tag{7.2}$$

After this, notice that $(x^n q_n \,-\, q_{n-1}x^{n+1}) = x^n$ which gives us

$$A_{n-1}(x)A_{n+1}(x) \,-\, 4(A_n(x))^2 \;=\; 4\left(4^{n^2}\right)^2 \left(3q_n^2 - 4x^n\right) . \tag{7.3}$$

Thus, it suffices to compute the coefficient $[x^n]\left(3q_n^2 - 4x^n\right)$: first we notice that the coefficient of $x^n$ in $q_n^2$ is $(n+1)$ which gives us $[x^n]\left(3q_n^2 - 4x^n\right) = 3n - 1$; $3n - 1$ is always nonnegative for $n \geq 1$.

$\underline{(+,+,+,-)}$: Here an example is the *Delannoy triangle*: consider the matrix $A$ with entries satisfying the linear recurrence

$$a_{n,k} \;=\; a_{n-1,k-1} \,+\, a_{n-1,k} \,+\, a_{n-1,k+1} \tag{7.4}$$

for $n \geq 1$, with initial conditions $a_{0,k} = \delta_{k0}$ and $a_{-1,k} = 0$. This is the matrix $T$ defined by the recurrence (4.1) under the specialization $a = d = f = 0$ and $c = e = g = 1$. This is also [21, A008288] written as a triangular array. It was shown in [7, Corollary 5.5] that this matrix has a totally positive left production matrix $Q$ which implies that the first three answers are true. Finally, notice that the Hankel minor $A_1(x)A_3(x) - (A_2(x))^2 = -x^2$ which is not coefficientwise positive.

$\underline{(+,-,-,+)}$: Here we will consider two examples considered by Zhu [36]:

(1) Our first example is his *third Catalan triangle* which was introduced by Radoux [23]

$$A = \left(\frac{2k+1}{2n+1}\binom{2n+1}{n-k}\right)_{n,k \geq 0} . \tag{7.5}$$

Zhu denoted this matrix as his Fuss–Catalan Riordan array $\mathcal{C}^{(2)}(1,2)$.

(2) Our second example is the *Ballot table* [21, A033184]: consider the matrix $\widehat{A}$ with entries

$$\widehat{A} \;=\; \left(\frac{k+1}{n+1}\binom{2n-k}{n-k}\right)_{n,k \geq 0} ; \tag{7.6}$$

this matrix was studied by Aigner [1]. Zhu denoted this matrix as his Fuss–Catalan Riordan array $\mathcal{C}^{(2)}(1,1)$.

For each of these examples, we first mention why questions (b), (c) are false, and also the truth values which directly follow from Zhu's work.

(1) We first mention that the minor $\overleftarrow{A}\left[\begin{array}{c}\{2,3,4,5,6\}\\ \{0,1,2,3,4\}\end{array}\right]$ is negative which shows that $\overleftarrow{A}$ is not TP. Also, we mention that $A_3(x) = x^3 + 5x^2 + 9x + 5$ has complex non-real roots. The total positivity of the matrix $A$ is a special case of [36, Theorem 6.2]. The coefficientwise Hankel total positivity of the row-generating polynomials is a special case of [36, Theorem 6.5].

(2) We first mention that the minor $\overleftarrow{\widehat{A}}\left[\begin{array}{c}\{1,2,3,4\}\\ \{0,1,2,3\}\end{array}\right]$ is negative which shows that $\overleftarrow{\widehat{A}}$ is not TP. Also, we mention that $\widehat{A}_3(x) = x^3 + 3x^2 + 5x + 5$ has complex non-real roots. The total positivity of the matrix $A$ is a special case of [36, Theorem 6.2].

It remains to show that for the second example, the sequence $(\widehat{A}_n(x))_{n\geq 0}$ is coefficientwise Hankel totally positive. To do this, we write out the following continued fraction identity:

**Proposition 7.1.** *The ordinary generating function of the row-generating polynomials of the ballot table has the following Jacobi-type continued fraction:*

$$\sum_{n=0}^{\infty} \widehat{A}_n(x)t^n \;=\; \cfrac{1}{1-(x+1)t-\cfrac{t^2}{1-2t-\cfrac{t^2}{1-2t-\cfrac{t^2}{1-2t-\dots}}}}\,. \tag{7.7}$$

*Therefore, the tridiagonal production matrix $T = (t_{n,k})_{n,k\geq 0}$ with entries*

$$t_{00} \;=\; x+1 \qquad t_{nn} \;=\; 2 \qquad t_{n+1,n} \;=\; 1 \qquad t_{n,n-1} \;=\; 1 \tag{7.8}$$

*when considered as a right production matrix gives as output sequence $(\widehat{A}_n(x))_{n\geq 0}$.*

Proof. First notice that the number $\widehat{a}_{n+1,k+1}$ can be written as $\widehat{a}_{n-1,k-1} = \dfrac{k}{2n-k}\binom{2n-k}{n}$; Deutsch [11, Section 6.6] showed that $\widehat{a}_{n-1,k-1}$ is the number of Dyck paths of semilength $n$ with $k$ returns to the $x$-axis.

Thus, using Flajolet's combinatorial interpretation of Stieltjes-type continued fractions [14], we see that the ordinary generating function of the sequence $1, x\widehat{A}_0(x), x\widehat{A}_1(x), x\widehat{A}_2(x), \dots$ is given by the following Stieltjes-type continued fraction expansion:

$$1 + xt\sum_{n=0}^{\infty} \widehat{A}_n(x)t^n \;=\; \cfrac{1}{1-\cfrac{xt}{1-\cfrac{t}{1-\cfrac{t}{1-\dots}}}}\,. \tag{7.9}$$

The desired identity (7.7) finally follows by using the odd contraction formula [30, pp. V-31—V-32]. $\square$

**IS THIS J-FRACTION KNOWN????????**

With the continued fraction/right production matrix in Proposition 7.1, we now prove the following statement:

**Proposition 7.2.** *The matrix $T$ defined in (7.8) is totally positive. As a consequence the sequence $(A_n(x))_{n\ge 0}$ is coefficientwise Hankel totally positive.*

Proof. Let $L$ be the lower bidiagonal matrix with entries $L_{n,n} = 1, L_{n,n-1} = 1$ and let $U$ be the upper bidiagonal matrix with entries $U_{0,0} = x$, $U_{n,n} = 1$ for $n \ge 1$, and $U_{n,n+1} = 1$. Then we can easily notice that the matrix $T$ has the factorization $T = UL$. As both matrices $L, U$ are totally positive, this implies that the matrix $T$ is also totally positive.

From Proposition 7.1, we obtain that the output sequence of the right production matrix $T$ is the sequence $(\widehat{A}_n(x))_{n\ge 0}$; as $T$ is totally positive, we get from [22, Theorem 9.7] that the output sequence is coefficientwise Hankel totally positive. □

**Remark.** Zhu in [36, Theorem 6.5] provides the following sufficient condition on the quantities $m, s, r$ for the row-generating polynomial sequence of the Fuss–Catalan Riordan array $C^{(m)}(s, r)$ to be totally positive: $r = m$ and $0 \le s \le m$. However, for the ballot table $A$, we have $m = 2, s = r = 1$ which does not fit this sufficient condition, but nevertheless, we still get coefficientwise Hankel total positivity of its row-generating polynomials. It will be interesting to find out the full necessary and sufficient condition on the triple of parameters $(m, s, r)$. ■

$\underline{(+,-,+,-)}$: Let $A$ be the lower bidiagonal matrix with diagonal entries $a_{n,n} = 1$ and subdiagonal entries given by $a_{n,n-1} = c_n$ for a sequence $(c_n)_{n\ge 1}$. If each $c_n \ge 0$ then this matrix is known to be totally positive. Next, notice that the row-generating function $A_n(x) = c_n x^{n-1} + x^n = (c_n + x)x^{n-1}$ which is also negative real-rooted whenever $c_n \ge 0$. However, for some indices $i, j$ with $i < j$, the minor $\overleftarrow{\widehat{A}}\begin{bmatrix} \{i,j\} \\ \{0,1\} \end{bmatrix}$ is $c_j - c_i$, which is negative if we further insist $c_i > c_j$. Finally, notice that

$$A_0(x)A_2(x) - A_1(x)^2 \;=\; (c_2 - 2c_1)x - c_1^2 \tag{7.10}$$

which is not coefficientwise positive if $c_1 > 0$.

$\underline{(+,+,-,-)}$: Let $A$ be the lower triangular matrix of all ones. Clearly, $A = \overleftarrow{A}$, and it is well known that this matrix is totally positive. The row-generating polynomials $A_n(x) = \sum_{k=0}^{n} x^k$ are also well known to have complex non-real roots whenever $n \ge 3$. Finally, notice that

$$A_0(x)A_2(x) - A_1(x)^2 \;=\; -x \tag{7.11}$$

and hence, the sequence $(A_n(x))_{n\ge 0}$ is not coefficientwise Hankel totally positive.

$\underline{(+,-,-,-)}$: Consider the following matrix $M$:

$$M \;=\; \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 2 & 1 & \\ 0 & 1 & 1 & 1 \end{bmatrix} \tag{7.12}$$

and construct $A = \left[\begin{array}{c|c} M & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{0} \end{array}\right]$. We only need to perform a small finite number of computations to check each assertion, and carrying these out establishes that $A$ is indeed an example of this class.

$\underline{(-,-,+,-)}$: Consider the following matrix $M$:

$$M = \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 100 & 1 & \\ 32 & 32 & 10 & 1 \end{bmatrix} \tag{7.13}$$

and construct $A = \left[\begin{array}{c|c} M & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{0} \end{array}\right]$. We only need to perform a small finite number of computations to check each assertion, and carrying these out establishes that $A$ is indeed an example of this class.

It remains to construct examples for the classes $(+,-,+,+), (-,-,+,+), (-,-,-,+)$. We will construct examples for these classes based on the properties of the famous Chebyshev polynomials; we do this in Section 7.1.

## 7.1 Examples based upon properties of Chebyshev polynomials

We first recall the definition of Chebyshev polynomials and mention some of their properties. The Chebyshev polynomials $T_n(x)$ are given by the three-term recurrence relationship

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \qquad \text{for } n \geq 1 \tag{7.14}$$

with initial conditions $T_0(x) = 1, T_1(x) = x$. They can also be rewritten as

$$T_n(x) = \cos(n \arccos(x)). \tag{7.15}$$

Either of the properties implies that $T_n(x)$ has degree $n$, and all its zeros lie in $(-1, 1)$. From the identity (7.19), it additionally follows that, on the interval $[-1, 1]$, the polynomial $T_n(x)$ for each $n$ attains $\left\lfloor \frac{n+1}{2} \right\rfloor$ times its maximal value $+1$ and $\left\lfloor \frac{n}{2} \right\rfloor$ times its minimal value $-1$; it also attains each value in $(-1, 1)$ exactly $n$ times. The three-term relation (7.14) can be rewritten as

$$T_{n+1}(2x+1) = (4x+2)T_n(2x+1) - T_{n-1}(2x+1) \tag{7.16}$$

with initial conditions $T_0(2x+1) = 1, T_1(2x+1) = 2x+1$.

We will require the following lemma:

**Lemma 7.3.** *Let $h_n(x)$ be a polynomial with degree at most $n$ with a positive leading coefficient, and let $g_n(x)$ be a polynomial defined by*

$$g_n(x) := T_n(2x+1) + h_n(x) . \tag{7.17}$$

*Further assume that $h_n$ satisfies $h_n([-1, 1]) \subseteq [-1, 1]$, such that $g_n(x)$ is also a polynomial of degree $n$. Then every zero $x_*$ of $g_n(x)$ lies in $[-1, 1]$; in fact, $x_* \in \{-1, 1\}$ is only possible when $h_n(x_*) = -1$ or $h_n(x_*) = 1$. Moreover, a zero $x_*$ can only be a multiple zero if $h_n(x_*) = \pm 1$ and $|x_*| < 1$.*

## 7.2 CONTENT FROM ALEX'S WRITEUP!!!!!!!!! INCORPORATE THIS SOON: Illustrations to that four positivity properties may occur in any combination

To each sequence of polynomials $\mathfrak{p} = \big(p_n(x)\big)_{n=0}^{\infty}$ such that $\deg p_n(x) = n$, we put into correspondence a lower-triangular matrix $A = \big(a_{nk}\big)_{n=0}^{\infty}$ of coefficients of the polynomials according to the rule

$$a_{nk} = [x^k]\, p_n(x).$$

Then the matrix $A^{rev} = \big(a_{n,n-k}\big)_{n=0}^{\infty}$ is built from the reversed coefficients of $\big(p_n(x)\big)_{n=0}^{\infty}$ or, equivalently, from coefficients of the sequence $\big(x^n p_n\big(\frac{1}{x}\big)\big)_{n=0}^{\infty}$ of reciprocal (i.e. reflected) polynomials.

The aim of the present note is to provide examples to the assertion that all combinations of presence/absence of the following positivity properties are possible:

- total positivity of the matrix $A$,
- total positivity of the matrix $A^{rev}$,
- infinite Toeplitz matrices $T(p_n)$ are totally positive for every $n$,
- Hankel total positivity of the sequence $p_n(x)$, i.e. total positivity of the Hankel matrix $H(\mathfrak{p}) = \big(p_{n+k}(x)\big)_{n,k=0}^{\infty}$.

The property (7.2) is equivalent to that all zeros of the polynomials $p_n(x)$ lie on the negative real semi-axis $(-\infty, 0]$ (plus positivity of their leading coefficients). The property (7.2) means that, for each $x > 0$ there exists a positive measure $\mu_x$ supported on $[0, \infty)$ whose $n$th moment is

$$p_n(x) = \int_0^{\infty} s^n d\mu_x(s), \qquad n = 0, 1, \dots . \tag{7.18}$$

The property (7.2) in pointwise sense is also equivalent to positive definiteness of two infinite matrices: $H(\mathfrak{p})$ and $H^1(\mathfrak{p}) = \big(p_{n+k+1}(x)\big)_{n,k=0}^{\infty}$.

# One example relying on Katkova–Vishnyakova theorem

The Katkova–Vishnyakova theorem asserts that a sequence growing fast enough must be Hankel totally positive. Namely, if $p_{n-1}(x)p_{n+1}(x) \geq 4p_n^2(x)$ for all $n = 1, 2, \dots$, then the Hankel matrix $\big(p_{n+k}(x)\big)_{n,k=0}^{\infty}$ is totally positive. As is shown by A. Sokal, a coefficientwise extension of this theorem with similar conditions also holds true.

We are interested in the example for [**the case** $(+,+,-,+)$ — **(c)**]. Observe that the polynomials

$$p_n(x) = 4^{n^2} \sum_{k=0}^{n} (x+1)^k$$

are not Toeplitz totally positive: none of the polynomials for $n \geq 2$ is real-rooted (straightforward: all coefficients of $p_n(x-1) = 4^{n^2} \sum_{k=0}^{n} x^k$ form a sequence of ones, which lacks the necessary condition for real-rootedness — log-concavity). At the same time, the corresponding matrix $A = A^{rev}$ is totally positive, since it coincides up to scaling of rows with the totally positive lower-triangular matrix of ones. Now, for brevity's sake denote $q_n = \sum_{k=0}^{n}(x+1)^k$; then

$$\begin{aligned} p_{n-1}(x)p_{n+1}(x) &= 4^{(n-1)^2}4^{(n+1)^2}q_{n-1}q_{n+1} = 16\cdot 4^{2n^2}q_{n-1}(q_n+(x+1)^{n+1}) \\ &= 16\cdot\left(4^{n^2}\right)^2\left(q_n^2 - q_n(x+1)^n + q_{n-1}(x+1)^{n+1}\right) \\ &= 4p_n^2(x) + 4\left(4^{n^2}\right)^2\left(3q_n^2 - 4q_n(x+1)^n + 4q_{n-1}(x+1)^{n+1}\right) \end{aligned}$$

However, the expression in parentheses allows a straightforward estimate

$$\begin{aligned} Q &= 3q_n^2 - 4q_n(x+1)^n + 4q_{n-1}(x+1)^{n+1} \\ &= 3q_{n-1}q_n + 3q_n(x+1)^n - 4q_n(x+1)^n + 4q_{n-1}(x+1)^{n+1} \\ &= 3q_{n-1}^2 + 3q_{n-1}(x+1)^n - q_n(x+1)^n + 4(x+1)^{2n} + 4q_{n-2}(x+1)^{n+1} \\ &= 3q_{n-1}^2 + 3q_{n-1}(x+1)^n - q_{n-1}(x+1)^n - (x+1)^{2n} + 4(x+1)^{2n} + 4q_{n-2}(x+1)^{n+1} \\ &= 3q_{n-1}^2 + 2q_{n-1}(x+1)^n + 3(x+1)^{2n} + 4q_{n-2}(x+1)^{n+1} \geq 0, \end{aligned}$$

the inequality here is in the coefficientwise sense (as well as for all $x \geq 0$). Thus,

$$p_{n-1}(x)p_{n+1}(x) \geq 4p_n^2(x) + 4\left(4^{n^2}\right)^2 Q \geq 4p_n^2(x),$$

whence the Hankel total positivity of $\big(p_n(x)\big)_{n=0}^{\infty}$ follows by the Katkova–Vishnyakova theorem.

## Examples based upon properties of Chebyshev polynomials

Recall that the Chebyshev polynomials $T_n(x)$ satisfy the three-term recurrence relationship

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x), \qquad T_0(x) = 1,\ T_1(x) = x$$

and can be rewritten as

$$T_n(x) = \cos(n \arccos(x)). \tag{7.19}$$

Either of the properties implies that $T_n(x)$ has degree $n$, and all its zeros lie in $(-1,1)$. From the identity (7.19), it additionally follows that, on the interval $[-1,1]$, the polynomial $T_n(x)$ for each $n$ attains $\lfloor\frac{n+1}{2}\rfloor$ times its maximal value $+1$ and $\lfloor\frac{n}{2}\rfloor$ times its minimal value $-1$; it also attains each value in $(-1,1)$ exactly $n$ times. The three-term relation can be rewritten as

$$T_{n+1}(2x+1) = (4x+1)T_n(2x+1) - T_{n-1}(2x+1), \qquad T_0(2x+1) = 1,\ T_1(2x+1) = 2x+1. \tag{7.20}$$

**Lemma 7.4.** *Let a polynomial $\phi_n$ of degree at most $n$ with a positive leading coefficient satisfy $-1 \le \phi_n(x) \le 1$ for all $-1 \le x \le 1$, so that*

$$g_n(x) \coloneqq T_n(2x+1) + \phi_n(x)$$

*is a polynomial of degree $n$. Then every zero $x_*$ of $g_n(x)$ lies in $[-1,1]$; in fact, $x_* \in \{-1,1\}$ is only possible when $\phi_n(-1) = -1$ or $\phi_n(-1) = -1$. Moreover, a zero $x_*$ can only be multiple if $\phi_n(x_*) = \pm 1$ and $|x_*| < 1$.*

Proof. The reasoning follows from a simple topological idea, that the zeros of $g_n$ are the points $x$ where the graphs of $T_n(2x+1)$ and $-\phi_n(x)$ intersect. The graph of $-\phi_n(x)$ is a curve connecting $-\phi_n(-1)$ with $-\phi_n(1)$ and lying in the rectangle $(-1,0)\times(-1,1)$. The graph of $T_n(2x+1)$ crosses the same rectangle $\lfloor \frac{n}{2} \rfloor$ times downwards and $\lceil \frac{n}{2} \rceil$ times upwards, so that $-\phi_n(x)$ must meet $T_n(2x+1)$ at least $n$ times — excluding the case $\phi_n(x_*) = T_n(2x_*-1) = \pm 1$ for $|x_*| < 1$, where $T_n'(2x_*-1) = 0 \ne T_n''(2x_*-1)$; if so, then, since the graph of $-\phi_n(x)$ remains in the rectangle, necessarily $\phi_n'(x_*) = 0$ and $\phi_n''(x_*) \cdot T_n''(2x_*-1) \ge 0$. The latter means, that $g_n$ has a double zero at $x_*$. To conclude: $g_n(x) = T_n(2x+1) + \phi_n(x)$ has precisely $n$ zeros in the interval $(-1,0)$ counting each zero so many times, which multiplicity it has.

The examples being sough rely on the following three facts.

**Theorem 7.5** (Schoenberg)**.** *If the sequence of orthogonal polynomials with positive leading coefficients corresponds to a measure supported on $(-\infty, 0]$, then the corresponding lower-triangular matrix of the coefficient is totally positive.*

**Remark**

1. The measure whose moments are the shifted Chebyshev polynomials $T_n(2x+1)$ is supported on two points:

$$1 + 2x - 2\sqrt{x+x^2} \quad \text{and} \quad 1 + 2x + 2\sqrt{x+x^2},$$

so that

$$T_n(2x+1) = \int s^n d\mu_x(s) = \frac{\left(1+2x+2\sqrt{x^2+x}\right)^n}{2} + \frac{\left(1+2x-2\sqrt{x^2+x}\right)^n}{2}.$$

2. If $\mu_x$ and $\nu_x$ are two measures as in (7.18), then

- their moments necessarily have positive leading coefficient in $x$ (for proving it is enough to take $x$ sufficiently large), and
- for each $n$, the $n$th moments of the positive measures $\mu_x + \nu_x$, $\mu_x + \nu_1$ and $\mu_x + \frac{1}{2}(\nu_x + \nu_1)$ are also polynomials in $x$ of degree $n$.

■ In particular, if $\nu_x(s) = \delta(s-x)$ stands for the unit point mass at $x$, then its moments are $\left(x^n\right)_{n=0}^{\infty}$; therefore,

$$\Big(T_n(2x+1) + x^n\Big)_{n=0}^{\infty}, \quad \Big(T_n(2x+1) + 1\Big)_{n=0}^{\infty} \quad \text{and} \quad \Big(T_n(2x+1) + \frac{1}{2} + \frac{1}{2}x^n\Big)_{n=0}^{\infty}$$

are Hankel totally positive sequences, and all zeros of the listed polynomials lie in $(-1,0)$. In fact, the former two sequences are moments of measures with three point masses, while the latter sequence is built of a measure with four point masses.

**The case** $(+,-,+,+)$ — **(b)** By Schoenberg's Theorem, the matrix $\widetilde{A}$ built from the sequence $\big(T_n(2x+1)\big)_{n=0}^{\infty}$ is totally positive; thus, the matrix $A$ equal to $\widetilde{A}$ with the first column multiplied by 2 is also totally positive. But this $A$ (due to $T_n(1)=1$) is built from the sequence $\big(T_n(2x+1)+1\big)_{n=0}^{\infty}$. At the same time, its reversal $A^{rev}$ (which coincides with the matrix $A$ for $\big(T_n(2x+1)+x^n\big)_{n=0}^{\infty}$) is not totally positive:

$$A^{rev}\begin{pmatrix}1,2,\dots,10\\0,1,\dots,9\end{pmatrix} = -767389143067603842836058075 < 0;$$

here and below the indexation of rows and columns starts with 0.

**The case** $(-,-,+,+)$ – **(e)** Now, we construct $A$ for the sequence $\big(T_n(2x+1)+\frac{1}{2}+\frac{1}{2}x^n\big)_{n=0}^{\infty}$, in which case

$$A\begin{pmatrix}3,4,\dots,27\\0,1,\dots,24\end{pmatrix} = -510710909.19654\ldots\times 10^{200} < 0.$$

Simultaneously,

$$A^{rev}\begin{pmatrix}1,2,3,4,5\\0,1,2,3,4\end{pmatrix} = -\frac{95835}{32} < 0.$$

**The case** $(-,-,-,+)$ — **(k)** On account that $2+x^n>1$ on the interval $(-1,0)$, and that the slope of $T_n(2x+1)$ for $n\geq 2$ is steeper than the slope of $2+x^n$ on $(-\infty,-1)\cup(0,+\infty)$, the polynomials $T_n(2x+1)+2+x^n$ cannot have more than two real zeros (and those zeros can only be simple). In other words, the coefficients of $T_n(2x+1)+2+x^n$ for each $n>2$ are not Toeplitz totally positive. The other properties are analogous to the case (e). The negative minors of $A$ and $A^{rev}$ here are, e.g.:

$$A\begin{pmatrix}1,2,\dots,10\\0,1,\dots,9\end{pmatrix} = -2302167429202811528508174225 < 0$$

and

$$A^{rev}\begin{pmatrix}1,2\\0,1\end{pmatrix} = \begin{vmatrix}3 & 3\\9 & 8\end{vmatrix} = -3 < 0.$$

**The case** $(+,-,-,+)$ — **(f)** It is clear from the graphs of polynomials $T_n(2x+1)+2$, that they have no real zeros for even $n$, and exactly one real zero for odd $n$. Moreover, that zero for odd $n$ is simple. The rest is similar to the case (b), the negative minor here is

$$A^{rev}\begin{pmatrix}1,2,3,4,5\\0,1,2,3,4\end{pmatrix} = -1160240 < 0.$$

The deficiency of the listed examples is that the corresponding Hankel matrices may lack coefficientwise total positivity, they must only be pointwise totally positive.

For instance, some of the coefficients of the polynomial

$$\det\left(T_{n+k}(2x+1)+x^{n+k}\right)_{n,k=0}^{2}$$
$$=\det\begin{pmatrix} 2 & 3x+1 & 9x^2+8x+1 \\ 3x+1 & 9x^2+8x+1 & 33x^3+48x^2+18x+1 \\ 9x^2+8x+1 & 33x^3+48x^2+18x+1 & 129x^4+256x^3+160x^2+32x+1 \end{pmatrix}$$
$$=4x(x+1)^3(3x-1)^2=4x-12x^2-24x^3+40x^4+84x^5+36x^6$$

are negative. Analogously,

$$\det\left(T_{n+k+1}(2x+1)+\frac{1}{2}+\frac{1}{2}x^{n+k+1}\right)_{n,k=0}^{2}=3x^2+54x^3+86x^4-32x^5-97x^6+42x^7+72x^8.$$

# Alternative example for the case $(-,?,+,+)$ — (e) or reversed (b):

Direct verification of coefficientwise Hankel positivity is a complicated task. One can take a sequence induced by an S-continued fraction with coefficientwise positive coefficients; this option, however, only allows constructing sequences of polynomials, whose triangular matrices of coefficients are coefficientwise totally positive. Another option is to employ certain other types of continued fractions — such as T-fractions. Absence of the TP property (7.2) of the corresponding matrix $A$ may be achieved via choosing certain unusual pattern with respect to $x$ in the fraction's coefficients. For example, consider the T-fraction

$$F=\cfrac{1}{1-xy^2t-\cfrac{(x+1)t}{1-yt-\cfrac{2(x+4)t}{1-4xyt}}},$$

which can be rewritten as

$$F=\frac{-4t^2xy^2+t(4xy+2x+y+8)-1}{4x^2y^4t^3-xy(4xy^2+2xy+4x+y^2+12y+4)t^2+(xy^2+4xy+3x+y+9)t-1}$$
$$=\sum_{n=0}^{\infty}p_n(x;y)t^n.$$

The polynomials $p_n$ are the Thron–Rogers polynomials generated by $F$, the first few of them are

$$\begin{aligned} p_0(x;y)&=1,\\ p_1(x;y)&=1+x(1+y^2),\\ p_2(x;y)&=(9+y)+x(12+y+2y^2)+x^2(3+2y^2+y^4),\\ p_3(x;y)&=(81+18y+y^2)+x(135+56y+20y^2+2y^3)\\ &\quad+x^2(63+46y+26y^2+2y^3+3y^4)+x^3(9+8y+7y^2+3y^4+y^6). \end{aligned}$$

It is known from the general theory that the Thron–Rogers polynomials form a Hankel totally positive sequence (in the coefficientwise sense).

Since the denominator of $F$ has degree 3, the whole sequence $(p_n)_{n=0}^{\infty}$ satisfies the four-term relations

$$\begin{aligned} p_{n+1}(x;y) = {} & \big((y^2+4y+3)+y+9\big)\, p_n(x;y) \\ & - xy\,\big((4y^2+2y+4)x+y^2+12y+4\big)\, p_{n-1}(x;y) + 4x^2y^4\, p_{n-2}(x;y), \end{aligned} \tag{7.21}$$

whose coefficients do not depend on $n$. One can additionally introduce

$$p_{-1}(x;y) = \frac{1}{xy^2},$$

so that the recurrence (7.21) remains true for $n = 1$. Clear that the coefficients of $p_n(x;y)$ in $x$ are coefficientwise positive polynomials in $y$. However, the matrix $A$ built of coefficients $(p_n(x;y))_{n=0}^{\infty}$ is not totally positive even in the pointwise sense. Indeed, letting, for instance, $y = 1/5$ yields

$$A\binom{1,2,3,4}{0,1,2,3} = -\frac{28946026683516}{30517578125} < 0.$$

So far I was unable to find a negative minor in the matrix $A^{rev}$. **What if that matrix is indeed totally positive?** The four-term recurrence relation (7.21) for the reciprocal polynomials $q_n = x^n p_n\big(\frac{1}{x}, y\big)$ yields

$$\begin{aligned} q_{n+1} = {} & \big(y^2+4y+3+(y+9)x\big)\, q_n \\ & - y\,\big(4y^2+2y+4+\big(y^2+12y+4\big)\,x\big)\, q_{n-1} + 4xy^4\, q_{n-2} \end{aligned}$$

with the initial data

$$q_{-1} = q_0 = 1, \qquad q_1 = 1+y^2+x.$$

The solution of the third-order difference equation (7.21) is determined by the roots of the characteristic equation

$$4x^2y^4t^3 - xy(4xy^2+2xy+4x+y^2+12y+4)t^2 + (xy^2+4xy+3x+y+9)t - 1 = 0 \tag{7.22}$$

in $t$. This equation is cubic; let $\Phi_1, \Phi_2, \Phi_3$ denote its roots. Unless some of these roots coincide, the general solution of (7.21) can be written as

$$p_n(x;y) = C_1\Phi_1^n + C_2\Phi_2^n + C_3\Phi_3^n. \tag{7.23}$$

Although $C_1, C_2, C_3$ and $\Phi_1, \Phi_2, \Phi_3$ depend on both $x$ and $y$, they are independent of $n$.

Note that the left-hand side of (7.22) vanishes on substituting $x = -1$ and $t = -y^{-2}$. So, due to $p_{-1}(-1;y) = -y^{-2}$, $p_0(-1;y) = 1$ and $p_1(-1;y) = -y^2$, we obtain

$$p_n(-1;y) = (-1)^n y^{2n}. \tag{7.24}$$

For our example we fix $y = \frac{1}{5}$. The coefficients[3] of (7.22) are real for real $x$, so at least one of the roots $\Phi_1, \Phi_2, \Phi_3$ is real; the remaining two zeros are complex or real depending on whether the discriminant

$$\frac{4x^2}{9765625}\left(22535064x^4 + 194111748x^3 + 570837474x^2 + 693156635x + 294137225\right)$$

of (7.22) is positive or negative. The zeros of the discriminant are

$$\begin{aligned} x_1 &= -3.997733829\dots, \quad x_2 = -1.914987165\dots, \\ x_3 &= -1.695428699\dots, \quad x_4 = -1.005615414\dots, \quad x_{5,6} = 0, \end{aligned}$$

and the negativity intervals are $(x_1, x_2)$ and $(x_3, x_4)$. The zeros $x_{5,6}$ stand for the difference equation (7.21) becoming linear, and hence

$$p_n(0; 1/5) = C_1\Phi_1^n \quad \text{with} \quad \Phi_1 = \frac{5}{46}.$$

The other four zeros of the discriminant correspond to that there is a (unique) multiple zero: say, $\Phi_2 = \Phi_3$, in which case (7.23) turns into

$$p_n(x; y) = C_1\Phi_1^n + C_2\Phi_2^n + C_3 n\Phi_3^n. \tag{7.25}$$

Finally, observe that the only zero of $p_1(x; 1/5) = 0$ is $x_7 = -\frac{25}{26}$, while $p_2(x; 1/5)$ has two zeros $x_{8,9} = 5 \cdot \frac{-1535 \pm \sqrt{584305}}{3852}$.

**Theorem 7.6.** *Each of the polynomials $p_n(x; 1/5)$ for $n \ge 1$ is real-rooted. Moreover, all its zeros are simple and (except for the largest one) lie in the interval $(x_1, x_2)$. The largest zero lies in the interval*

$$[x_8, x_7] = [-1.0002616\dots, -0.9615384\dots];$$

*it oscillates near $-1$ and tends to $-1$ as $n \to \infty$.*

It seems possible to prove that the zeros of $p_n(x; 1/5)$ and $p_{n+1}(x; 1/5)$ in the interval $(x_1, x_2)$ are interlacing; then the interlacing property for $p_{2n-1}(x; 1/5)$ and $p_{2n}(x; 1/5)$ on the whole real line will only fail due to their largest zeros, $n = 1, 2, \dots$.

The signs of the coefficients of (7.21) allow a Sturm-like proof of the asserted zero localisation only for $x_1 < x < -115/48 = -2.395833\dots$; a modification of the same idea can be employed to cover a wider interval, but I have no idea how to extend it to the whole interval $(x_1, x_2)$. In contrast, the formula (7.25) here does the job.

Proof. Hereinafter $p_n(x)$ stands for $p_n(x; 1/5)$. The coefficients of (7.21) for $y = 1/5$ are

$$\frac{96x}{25} + \frac{46}{5}, \quad -\left(\frac{114x}{125} + \frac{161}{125}\right)x \quad \text{and} \quad \frac{4x^2}{625}.$$

They are *positive* for $x \in \left(-\frac{161}{114}, 0\right)$, the second one vanishes when $x = -\frac{161}{114}$. By induction: $p_0(x), p_2(x) > 0$ and $p_1(x) \ge 0$ for $x \ge x_7$, and hence $p_n(x) > 0$ for $x \ge x_7$

[3] If needed, these (rational) coefficients can be made integer through a proper scaling without breaking the example.

for all $n \geq 2$. On the other hand, $p_1(x_8) < 0$, $p_2(x_8) = 0$ and $p_3(x_8) < 0$, thus also $p_n(x_8) < 0$ for all $n \geq 3$. Thus each of the polynomials $p_n(x)$ has at least one zero in the interval $[x_8, x_7]$. From (7.24) we have

$$p_n(-1) = (-25)^{-n},$$

and hence consecutive polynomials have odd number of zeros in $[x_8, x_7]$ on different sides from the point $-1$. Furthermore, if $p_{n-1}(x_*) = 0$ for some $n$ and $x_* \in [x_8, x_7]$, while $p_{n-2}(x_*) \cdot p_n(x_*) > 0$, then positivity of coefficients of (7.21) implies that $p_n(x_*) \cdot p_{n+1}(x_*) > 0$; in other words,

$$\begin{aligned} x_* \in [x_8, -1) &\implies n \text{ is odd and } \exists x_\dagger \in (x_*, -1):\ p_{n+1}(x_\dagger) = 0;\\ x_* \in (-1, x_7] &\implies n \text{ is even and } \exists x_\dagger \in (-1, x_*):\ p_{n+1}(x_\dagger) = 0. \end{aligned}$$

To finish with the properties of the largest zero observe that

$$\begin{aligned} \Phi_1|_{x=-1} &= -\frac{1}{25} = -0.04,\\ \Phi_2|_{x=-1} &= \frac{1}{10}\left(27 - \sqrt{745}\right) = -0.029\ldots,\\ \Phi_3|_{x=-1} &= \frac{1}{10}\left(27 + \sqrt{745}\right) = 5.429\ldots, \end{aligned}$$

along with that $C_1|_{x=-1} = 1$ and $C_2|_{x=-1} = C_3|_{x=-1} = 0$, which are the isolated zeros[4] of $C_1, C_2$. So, from (7.25) one sees, that $p_n(x) \to C_3\Phi_3^n$ as $n \to \infty$ near $-1$, except for the point $x = -1$ corresponding to $p_n(-1) = \Phi_1^n$.

Indeed, let $x \in (x_1, x_2)$. Then the discriminant of (7.22) is negative, and hence there is one real solution of (7.22) (denoted by $\Phi_1$) and a pair of complex conjugated solutions (respectively, $\Phi_2$ and $\Phi_3 = \overline{\Phi}_2$). Vieta's formulae then read

$$\Phi_1 + \Phi_2 + \Phi_3 = \frac{96x}{25} + \frac{46}{5}; \tag{7.26a}$$

$$\Phi_1\Phi_2 + \Phi_1\Phi_3 + \Phi_2\Phi_3 = \left(\frac{114x}{125} + \frac{161}{125}\right)x; \tag{7.26b}$$

$$\Phi_1\Phi_2\Phi_3 = \frac{4x^2}{625}. \tag{7.26c}$$

The last one implies that $\Phi_2 \neq 0$ and

$$\Phi_1 = \frac{4x^2}{625|\Phi_2|^2} > 0. \tag{7.27}$$

In fact, (7.26c) additionally shows that all solutions $\Phi_1, \Phi_2, \Phi_3$ are uniformly separated from the origin for $x$ varying on compact subsets of the punctured real line $\mathbb{R} \setminus \{0\}$; in particular, there exists a constant $\Phi_1^*$ such that $\Phi_1 > \Phi_1^* > 0$ whenever $x_1 \leq x \leq x_2$.

Our nearest aim is to show that the magnitude of $\Phi_1$ is small enough with respect to $|\Phi_2|$. This can be achieved by obtaining a proper lower estimate for $|\Phi_2|^2$. From (7.26a) one has

$$2\Phi_1 \operatorname{Re}\Phi_2 = \Phi_1\left(\frac{96x}{25} + \frac{46}{5}\right) - \Phi_1^2,$$

[4] The fact that the zeros are isolated can be shown strictly if needed.

Then (7.26b) can be rewritten as

$$|\Phi_2|^2 + 2\Phi_1 \operatorname{Re}\Phi_2 - \left(\frac{114x}{125} + \frac{161}{125}\right)x = 0,$$

and therefore

$$|\Phi_2|^2 - \left(\frac{114x}{125} + \frac{161}{125}\right)x + \Phi_1\left(\frac{96x}{25} + \frac{46}{5}\right) = \Phi_1^2,$$

or, due to (7.27),

$$|\Phi_2|^4 - \left(\frac{114x}{125} + \frac{161}{125}\right)x|\Phi_2|^2 + \frac{4x^2}{625}\left(\frac{96x}{25} + \frac{46}{5}\right) = \Phi_1^2|\Phi_2|^2 > 0. \tag{7.28}$$

The latter implies that either

$$|\Phi_2|^2 < \frac{x}{250}\left(161 + 114x + \sqrt{36x(361x + 977) + 22241}\right),$$

or

$$|\Phi_2|^2 > \frac{x}{250}\left(161 + 114x - \sqrt{36x(361x + 977) + 22241}\right). \tag{7.29}$$

Substitution of $x = -\frac{115}{48}$ (the point where the right-hand side of (7.26a) vanishes) into the former of the obtained inequalities yields $|\Phi_2| = 0$, which is impossible. However, the expression $36x(361x+977)+22241$ remains positive for all $x \in (x_1, x_2)$, so the continuity of $|\Phi_2|^2$ with respect to $x$ yields that only the latter inequality (7.29) can be true.

The estimate (7.29) shows that, in particular,[5]

$$|\Phi_2|^2 > 0.4310534203|x| > 0.82546$$

and, therefore,[6]

$$0 < \Phi_1 = \frac{4x^2}{625|\Phi_2|^2} < 0.0148473477|x| < 0.059355745.$$

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[5]The estimate for $|\Phi_2|^2$ can be made quadratic, e.g.:

$$\begin{aligned}|\Phi_2|^2 &> \frac{x}{250}\left(161 + 114x - \sqrt{36x(361x + 977) + 22241}\right)\\ &> \frac{x}{250}\left(161 + 114x - \sqrt{(119x + 177)^2}\right) = x\frac{338 + 233x}{250}.\end{aligned}$$

Analogously we can derive various upper bounds for $|\Phi_2|^2$: on, e.g., estimating (very roughly, one could use $\frac{x^2}{4489}$ instead) the value of $\Phi_1^2$ by $\frac{x^2}{125}$ in (7.28), we obtain

$$\begin{aligned}|\Phi_2|^2 &< \frac{x}{250}\left(161 + 115x - \sqrt{13225x^2 + 35494x + 22241x}\right)\\ &< x\frac{297 + 225x}{250} = \frac{9x(33 + 25x)}{250} < 9.648.\end{aligned}$$

[6]Curious: $0.0148473477 < \frac{1}{67}$, and the difference is below $10^{-4}$. However, to have rational numbers in right-hand sides, one could also take $|\Phi_2|^2 > \frac{25}{58}|x|$ leading to $\Phi_1 < \frac{232}{15625}|x|$ with $15625 = 5^6$.

On account of the last inequality, the following estimate of the real part of $\Phi_2$ is induced by (7.26a):

$$3.8251526523x + 9.2 < 2\operatorname{Re}\Phi_2 = 3.84x + 9.2 - \Phi_1 < 3.84x + 9.2. \tag{7.30}$$

Observe that $\operatorname{Re}\Phi_2$ changes its sign on the interval $-2.4052 < x < -2.3958(3) = -(115/48)$.

Without loss of generality, suppose that $\operatorname{Im}\Phi_2 > 0$, so that $\operatorname{Im}\Phi_3 = -\operatorname{Im}\Phi_2 < 0$. Recall that, $\Phi_2, \Phi_3$ change continuously and cannot become real for $x_1 < x < x_2$, as $x_1$ and $x_2$ are consecutive zeros of the discriminant. At the same time, if $x \to x_1 + 0$ or $x \to x_2 - 0$, the pair of complex conjugated zeros $\Phi_2, \Phi_3$ becomes a multiple zero on the real line, that is $\operatorname{Im}\Phi_2 \to +0$. Thus, the inequality (7.30) yields

$$0 < \arg\Phi_2 < \pi, \quad \lim_{x\to x_2-0} \arg\Phi_2 = 0, \quad \lim_{x\to x_1+0} \arg\Phi_2 = \pi. \tag{7.31}$$

Our next aim is to estimate the values of $C_1, C_2, C_3$. They solve the linear system

$$\begin{pmatrix} \Phi_1^{-1} & \Phi_2^{-1} & \Phi_3^{-1} \\ 1 & 1 & 1 \\ \Phi_1 & \Phi_2 & \Phi_3 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = \begin{pmatrix} p_{-1}(x) \\ p_0(x) \\ p_1(x) \end{pmatrix} = \begin{pmatrix} 25/x \\ 1 \\ \frac{26}{25}x + 1 \end{pmatrix}, \tag{7.32}$$

whose matrix does not degenerate. Therefore,

$$C_1 = \Phi_1 \frac{\Phi_2\Phi_3 \cdot \frac{25}{x} - (\Phi_2 + \Phi_3) + (\frac{26}{25}x + 1)}{(\Phi_1 - \Phi_2)(\Phi_1 - \Phi_3)}, \tag{7.33a}$$

$$C_2 = \Phi_2 \frac{\Phi_3\Phi_1 \cdot \frac{25}{x} - (\Phi_1 + \Phi_3) + (\frac{26}{25}x + 1)}{(\Phi_2 - \Phi_1)(\Phi_2 - \Phi_3)}, \tag{7.33b}$$

$$C_3 = \Phi_3 \frac{\Phi_1\Phi_2 \cdot \frac{25}{x} - (\Phi_1 + \Phi_2) + (\frac{26}{25}x + 1)}{(\Phi_3 - \Phi_1)(\Phi_3 - \Phi_2)}. \tag{7.33c}$$

It is immediately seen from here that $C_3 = \overline{C}_2$.

On account that $\Phi_1\Phi_2\Phi_3 \cdot \frac{25}{x} = \frac{4x}{25}$ and $\left(\frac{96x}{25} + \frac{46}{5}\right)\Phi_1 < 1.8464492849\Phi_1 < 0.0274148746|x|$, we have

$$\begin{aligned} &(\Phi_2 - \Phi_1)(\Phi_3 - \Phi_1) = |\Phi_2|^2 - (2\Phi_1 \operatorname{Re}\Phi_2 + \Phi_1^2) + 2\Phi_1^2 \\ &\quad = |\Phi_2|^2 - \Phi_1\left(\frac{96x}{25} + \frac{46}{5}\right) + 2\Phi_1^2 > |\Phi_2|^2 - 0.0274148746|x| > 0.4036385457|x|, \end{aligned}$$

and hence the coefficient $C_1$ is negative:

$$\begin{aligned} C_1 &= \Phi_1 \frac{\frac{25}{x}|\Phi_2|^2 + \frac{26}{25}x + 1 - 2\operatorname{Re}\Phi_2}{|\Phi_2 - \Phi_1|^2} = \frac{\frac{4x}{25} + \Phi_1\left(\frac{26}{25}x + 1 - \left(\frac{96x}{25} + \frac{46}{5} - \Phi_1\right)\right)}{|\Phi_2 - \Phi_1|^2} \\ &= \frac{\frac{4x}{25} - \Phi_1\left(\frac{14x}{5} + \frac{41}{5}\right) + \Phi_1^2}{|\Phi_2 - \Phi_1|^2} < \frac{\frac{4x}{25} + \Phi_1 \cdot 2.838035937 + \Phi_1 \cdot 0.059355745}{|\Phi_2 - \Phi_1|^2} < \frac{0.116981418x}{|\Phi_2 - \Phi_1|^2} < 0. \end{aligned}$$

At the same time,

$$\begin{aligned} C_1 = \frac{\frac{4x}{25} - \Phi_1\left(\frac{14x}{5} + \frac{41}{5}\right) + \Phi_1^2}{|\Phi_2 - \Phi_1|^2} > \frac{\frac{4x}{25} - 3\Phi_1}{|\Phi_2 - \Phi_1|^2} > \frac{0.2045420431x}{|\Phi_2 - \Phi_1|^2} > \frac{0.2045420431x}{0.4036385457|x|} \\ > -0.50674556551. \end{aligned}$$

As an immediate consequence, from the second row of (7.32) we have

$$1 < 2\operatorname{Re} C_2 = 2\operatorname{Re} C_3 = C_2 + C_3 = 1 - C_1 < 1.50674556551.$$

In its turn, the last equality implies that both $C_2$ and $C_3 = \overline{C}_2$ lie in the right-hand half-plane, i.e. $\arg C_2$ and $\arg C_3$ are in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Furthermore, the expression (7.33b) yields

$$\lim_{x\to x_2-0} \arg C_2 = \arg \frac{\frac{4x_2}{25} - \Phi_2(\Phi_1+\Phi_3) + \Phi_2\left(\frac{26}{25}x_2 + 1\right)}{(\Phi_2-\Phi_1)(\Phi_2-\Phi_3)} = \arg \frac{-2.3391637741\ldots}{\Phi_2-\Phi_3} = \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

and

$$\lim_{x\to x_1-0} \arg C_2 = \arg \frac{\frac{4x_1}{25} - \Phi_2(\Phi_1+\Phi_3) + \Phi_2\left(\frac{26}{25}x_1 + 1\right)}{(\Phi_2-\Phi_1)(\Phi_2-\Phi_3)} = \arg \frac{0.1198039604175453 57\ldots}{\Phi_2-\Phi_3} = -\frac{\pi}{2};$$

the respective equalities for $C_3$ follow on account that $\arg C_3 = -\arg C_2$.

Now we are in a position to complete the proof of the theorem. It is enough to suppose $n \geq 1$, as $p_0 \equiv 1$. The idea behind our reasoning is the same as in Lemma 7.4. The expression (7.25) with the notation

$$r = |\Phi_2| = |\Phi_3|, \quad \theta_j = \arg \Phi_j, \quad \eta_j = \arg C_j$$

reduces to

$$p_n\left(x; \frac{1}{5}\right) = C_1\Phi_1^n + |C_2|\, r^n \left(e^{i\eta}e^{in\theta} + e^{-i\eta}e^{-in\theta}\right) = C_1\Phi_1^n + 2\,|C_2|\, r^n \cos\left(n\theta + \eta\right). \tag{7.34}$$

Due to $|C_2| > \operatorname{Re} C_2 > \frac{1}{2}$, the first term on the right-hand side of (7.34) satisfies the estimate $|C_1\Phi_1^n| < 0.50674556551 \cdot 0.059355745^n$. It is small in relation to the amplitude of the remainder term

$$\left|\frac{C_2\Phi_2^n + C_3\Phi_3^n}{\cos\left(n\theta+\eta\right)}\right| = 2\,|C_2|\, r^n > r^n > 0.8254617674^n.$$

As $x$ runs from $x_1$ to $x_2$, the value of $\theta$ continuously changes from $-\pi$ to 0, while $\eta$ continuously changes from $\frac{\pi}{2}$ to $-\frac{\pi}{2}$. Hence, $n\theta + \eta$ runs between $-n\pi + \frac{\pi}{2}$ and $-\frac{\pi}{2}$, so that $\cos\left(n\theta+\eta\right)$ for $n \geq 1$ attains every value in $(0, 1)$ at least $n-1$ times. (No assumption on monotonicity of the arguments in $x$ is needed.) The zeros of $p_n(x; 1/5)$ are the points where

$$\cos\left(n\theta+\eta\right) = -C_1\Phi_1^n r^{-n}.$$

However, the graph of the left-hand side must meet at least $n-1$ times the graph of the right-hand side $-C_1\Phi_1^n r^{-n}$ lying in the strip $0 < -C_1(\Phi_1^*)^n r^{-n} < x < 0.03007$. So, on the one hand, $p_n(x; 1/5)$ has at least $n-1$ distinct zeros in the interval $(x_1, x_2)$. Moreover, $p_n(x; 1/5)$ has at least one zero outside $(x_1; x_2)$ near the point $-1$. On the other hand, $p_n(x; 1/5)$ cannot have more than $\deg p_n = n$ zeros, so all zeros of $p_n(x; 1/5)$ are real, simple and localised as asserted by the theorem.

# Tool and computational resource disclosure

The authors extensively used Mathematica to perform experiments related to the content of this paper. The authors declare that they *did not use AI assistance* to write this manuscript or to prove any of the results. However, on very few ocassions, AI tools were used to generate Mathematica code used in some of the experiments, or while searching for references. All mathematical statements, proofs, references, and final wording were checked and approved by the authors, who take full responsibility for the content of the manuscript.

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